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521. ON DR. WIENER'S MODEL OF A CUBIC SURFACE WITH 27 REAL LINES; AND ON THE CONSTRUCTION OF A DOUBLE-SIXER.

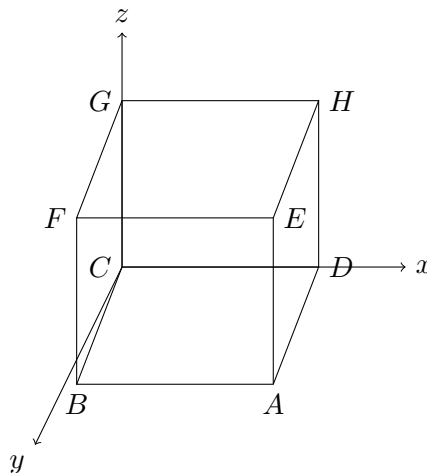
The model is a solid figure bounded by portions of the faces of the cube, and by a portion of the cubic surface, being a surface with three apertures, the collocation of which is not easily explained.

To determine the construction I measured, on the faces of the cube, the coordinates of the two extremities of each of the twelve lines; these were measured in tenths of an inch (taking account of the half division, or twentieth of an inch), and the resulting numbers divided by 16 to reduce them to the before-mentioned unit of 1.6 inches. These reduced values are shewn in the table: knowing then the coordinates of two points on each line, the equations of the several lines became calculable; the true theoretical form of these results—viz. the form which, but for errors of the model, or of the measurement, they would have assumed—is

$$\begin{aligned}
 b_1, \quad x &= B_1z + D, & y &= B'_1z + D', \\
 b_2, \quad x &= 0, & z &= 1, \\
 b_3, \quad x &= B_3(z + \beta_3), & y &= B'_3(z + \beta_3), \\
 b_4, \quad x &= B_4(z + \beta_4), & y &= B'_4(z + \beta_4), \\
 b_5, \quad y &= 0, & z &= -1, \\
 b_6, \quad x &= B_6(z + \beta_6), & y &= B'_6(z + \beta_6), \\
 a_1, \quad x &= 0, & y &= 0, \\
 a_2, \quad x &= A_2z + C_2, & y &= A'_2(z - 1), \\
 a_3, \quad x &= A_3(z + 1), & y &= A'_3(z - 1), \\
 a_4, \quad x &= A_4(z + 1), & y &= A'_4(z - 1), \\
 a_5, \quad x &= A_5(z + 1), & y &= A'_5z + C'_5, \\
 a_6, \quad x &= A_6(z + 1), & y &= A'_6(z - 1);
 \end{aligned}$$

but in consequence of such errors, the results are not accurately of the form in question.

The faces of the cube being as in the diagram:



the Table is as follows.

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Equations calculated from the measurements of the model, with
endpoint data on the faces of the cube.

Line	Calculated equation	First face data	Second face data	Additional face data
a_1	$x = 0, y = 0$	$ABCD : z = -5.688$	$EFGH : z = +5.688$	
a_2	$x = -0.780z - 0.187, y = -0.423z + 0.406$	$ABCD : x = 4.250, y = 0$	$EFGH : x = -4.625, y = 0$	
a_3	$x = -0.654z - 0.656, y = -0.588z + 0.531$	$ABCD : x = 3.062, y = 3.875$	$EFGH : x = -4.375, y = -2.812$	
a_4	$x = -2.912z - 2.959, y = -0.736z + 0.752$	$CDGH : z = 0.9375, y = -0.0625$	$AEDH : z = -2.969, y = 2.937$	
a_5	$x = 1.024z + 1.014, y = -1.049z - 0.277$	$ABCD : x = -4.812, y = 5.688$	$AEDH : y = -5.063, z = 4.562$	
a_6	$x = -0.264z - 0.187, y = -0.104z + 0.219$	$ABCD : x = -1.313, y = -0.8125$	$EFGH : x = 1.687, y = -0.375$	
b_1	$x = -1.611z + 0.151, y = -1.438z - 0.288$	$CDGH : y = 5.500, z = -3.625$	$AEDH : y = -4.656, z = 3.437$	
b_2	$x = 0, z = -1$	$AEBF : x = 0$	$BCFG : x = 0$	
b_3	$x = -1.352z - 0.685, y = -2.034z - 0.984$	$AEBF : z = 3.750, y = -3.281$	$BCFG : x = -3.812, z = 2.313$	
b_4	$x = -0.753z - 0.0315, y = -0.500z - 0.0315$	$ABCD : x = 4.250, y = 2.812$	$EFGH : x = -4.313, y = -2.875$	
b_5	$y = 0, z = 1$	$CDGH : x = 1, y = 0$	$AEDH : y = 0$	
b_6	$x = 1.123z - 0.702, y = -1.123z + 0.702$	$AEBF : x = -5.688, z = -4.438$	$BCFG : y = 0$	$AEDH : y = -5.688$

I hence calculate the intersections: considering any two lines which ought to intersect, then projecting on the horizontal plane and calculating x, y , the coordinates

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of the point of intersection of the two projections, these values of x, y substituted in the equations should give the same value of z ; but if the lines do not accurately intersect, then the values of z will be different.

	b_1	b_2	b_3	b_4	b_5	b_6
a_1	*	0, -1 0	0 0	0 0	0 0	0.398
a_2	+0.077 +.455	*	+0.381 +.771 .495 + .011	+4.292 +2.803 .052 + .010	-.967 -.008 + .008 +1	+.625 +.227 .673
a_3	-.129 + .013 .423	.001 + .001 -1	*	-4.782 -3.227 -5.704 + .036	+1 .028 + .028	+.346 + .076 +.344
a_4	+0.699 +.264 + .021	+1.119 -1	+1.286 +1.736	*	-5.871 +1 +.008 + .008	1.330 +.162 + .146
a_5	+0.957 +1.238 +.194	-.023 + .023 +1.488 .038 + .038	-1.398 + .060 +.282 +.045	+.412 .131 +.451	*	18.764 14.155
a_6	+.214 +.040 + .022	-1 +.323	+.283 .582 + .042	+.284 .423 + .208	-.057 + .057 -1	*

Starting from the assumed equations of $b_2, b_3, b_4, b_5, b_6, a_1$, and calculating by the theory the remaining lines, the equations of the b -lines (those of b_1 being calculated) are

$$\begin{aligned}
 b_1, \quad x &= 1.321z - .310, & y &= -1.295z + .581, \\
 b_2, \quad x &= 0, & z &= -1, \\
 b_3, \quad x &= -1.352(z + .510), & y &= -2.034(z + .510), \\
 b_4, \quad x &= -.753(z + .052), & y &= -.500(z + .052), \\
 b_5, \quad y &= 0, & z &= 1, \\
 b_6, \quad x &= 1.123(z - .624), & y &= -1.123(z - .624).
 \end{aligned}$$

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and the equations of the a -lines (those of all but a_1 being calculated) are

$$\begin{aligned}
 a_1, \quad x &= 0, & y &= 0, \\
 a_2, \quad x &= -.753z - .091, & y &= -.498(z - 1), \\
 a_3, \quad x &= -.609(z + 1), & y &= -.677(z - 1), \\
 a_4, \quad x &= -2.506(z + 1), & y &= -.841(z - 1), \\
 a_5, \quad x &= .874(z + 1), & y &= -.967z - .288, \\
 a_6, \quad x &= .170(z + 1), & y &= -.071(z - 1).
 \end{aligned}$$

and thence for the points of intersection the coordinates are

	b_1	b_2	b_3	b_4	b_5	b_6
a_1	*	-1 0	0 0	0 0 + .197	0 0	.336 0
a_2	-.170 +.446	*	-.510 -1 +.662 +.996	nearly parallel a_2, b_4	-.844 +1 0	+.624 +.336
a_3	+.105 -.515	-1 0	*	+1.31 -.262 -1.805	+1 0	-1.218 +.325 -.641
a_4	+.782 -.155 -1.071	+1.354 -1 0	+1.438 -1.574 +2.164	*	-5.012 +1 0	-.053 -1.259 +1.259
a_5	.573 +1.323 +3.189 -2.849 +2.649	-1 0 +.679	.702 +.259 +.291	.561 +.383 +.255	*	-6.410 +6.410 +6.333
a_6	+.241 +.041 +.417	-1 +.142	-.074 +.112	+.131 +.087	+.340 0 +1	*

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II.

I have in a paper “On the double-sixers of a cubic surface,” *Quart. Math. Journal*, t. x. (1870), pp. 58–71, [459], obtained analytical expressions for the twelve lines of a double-sixer, and also calculated numerical values, which however (as there remarked) did not come out convenient ones for the construction of a figure. A different mode of treatment since occurred to me, by means of the following equation of the cubic surface

$$\left(\frac{x}{\alpha'} - \frac{y}{\beta'} + \frac{z}{\gamma'} - \frac{w}{\delta'}\right) \left(\frac{xz}{\alpha\gamma} - \frac{yw}{\beta\delta}\right) - k \left(\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} - \frac{w}{\delta}\right) \left(\frac{xz}{\alpha'\gamma'} - \frac{yw}{\beta'\delta'}\right) = 0,$$

which as will appear is a very convenient one for the purpose. We in fact obtain at once eight lines of the double-sixer; viz. these are

$$\begin{array}{ll} 1. & x = 0, w = 0, \\ 3. & y = 0, z = 0, \\ 5. & \frac{x}{\alpha} - \frac{y}{\beta} = 0, \quad \frac{z}{\gamma} - \frac{w}{\delta} = 0, \\ 6. & \frac{x}{\alpha'} - \frac{y}{\beta'} = 0, \quad \frac{z}{\gamma'} - \frac{w}{\delta'} = 0, \\ 2'. & x = 0, y = 0, \\ 4'. & z = 0, w = 0, \\ 5'. & \frac{x}{\alpha'} - \frac{w}{\delta'} = 0, \quad \frac{y}{\beta'} - \frac{z}{\gamma'} = 0, \\ 6'. & \frac{x}{\alpha} - \frac{w}{\delta} = 0, \quad \frac{y}{\beta} - \frac{z}{\gamma} = 0. \end{array}$$

and also five lines not belonging to the double-sixer, viz.

$$\begin{array}{ll} 12. & x = 0, \quad \left(-\frac{y}{\beta'} + \frac{z}{\gamma'} - \frac{w}{\delta'}\right) \frac{1}{\beta\delta} - k \left(-\frac{y}{\beta} + \frac{z}{\gamma} - \frac{w}{\delta}\right) \frac{1}{\beta'\delta'} = 0, \\ 23. & y = 0, \quad \left(\frac{x}{\alpha} + \frac{z}{\gamma} - \frac{w}{\delta}\right) \frac{1}{\alpha\gamma} - k \left(\frac{x}{\alpha'} + \frac{z}{\gamma'} - \frac{w}{\delta'}\right) \frac{1}{\alpha'\gamma'} = 0, \\ 34. & z = 0, \quad \left(\frac{x}{\alpha'} - \frac{y}{\beta'} - \frac{w}{\delta'}\right) \frac{1}{\beta\delta} - k \left(\frac{x}{\alpha} - \frac{y}{\beta} - \frac{w}{\delta}\right) \frac{1}{\beta'\delta'} = 0, \\ 41. & w = 0, \quad \left(\frac{x}{\alpha'} - \frac{y}{\beta'} + \frac{z}{\gamma'}\right) \frac{1}{\alpha\gamma} - k \left(\frac{x}{\alpha} - \frac{y}{\beta} + \frac{z}{\gamma}\right) \frac{1}{\alpha'\gamma'} = 0, \\ 56. & \frac{x}{\alpha} - \frac{y}{\beta} + \frac{z}{\gamma} - \frac{w}{\delta} = 0, \quad \frac{x}{\alpha'} - \frac{y}{\beta'} + \frac{z}{\gamma'} - \frac{w}{\delta'} = 0. \end{array}$$

The remaining lines of the double-sixer are then easily determined; viz. the lines 3, 5, 6, and 12 are met by the line 2', and by a second line 1'; this, as a line meeting 3, 5, 6, will be given by equations of the form

$$x - \frac{\alpha}{\beta}y = \phi \left(\frac{\delta}{\gamma}z - w \right), \quad x - \frac{\alpha'}{\beta'}y = \phi \left(\frac{\delta'}{\gamma'}z - w \right),$$

and observing that these equations, writing therein $x = 0$, give

$$\frac{z}{\gamma} - \frac{w}{\delta} = -\frac{\alpha}{\beta\delta\phi}y, \quad \frac{z}{\gamma'} - \frac{w}{\delta'} = -\frac{\alpha'}{\beta'\delta'\phi}y,$$

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the condition of intersection with the line 12 gives

$$\phi = \frac{\alpha' - k\alpha}{\delta' - k\delta},$$

which is the value of ϕ in the foregoing equations; and to these we may join the resulting equation

$$y\phi\beta\beta'(\gamma\delta' - \gamma'\delta) = z\gamma\gamma'(\alpha\beta' - \alpha'\beta).$$

Proceeding in like manner for the lines 3', 2, 4, the equations for the remaining four lines of the double-sixer are

$$\begin{array}{ll} 2. \quad \phi = \frac{\alpha' - k\alpha}{\delta' - k\delta}, & 1'. \quad \phi = \frac{\alpha' - k\alpha}{\gamma' - k\gamma}, \\ y\phi\beta\beta'(\gamma\delta' - \gamma'\delta) = z\gamma\gamma'(\alpha\beta' - \alpha'\beta), & x\phi\alpha\alpha'(\gamma\delta' - \gamma'\delta) = w\delta\delta'(\alpha\beta' - \alpha'\beta), \\ y\phi\delta\delta'(\alpha\beta' - \alpha'\beta) = w\beta\beta'(\gamma\delta' - \gamma'\delta), & x\phi\gamma\gamma'(\alpha\beta' - \alpha'\beta) = z\alpha\alpha'(\gamma\delta' - \gamma'\delta), \\ 4. \quad \phi = \frac{\beta' - k\beta}{\gamma' - k\gamma}, & 3'. \quad \phi = \frac{\beta' - k\beta}{\delta' - k\delta}, \\ y\phi\beta\beta'(\gamma\delta' - \gamma'\delta) = z\gamma\gamma'(\alpha\beta' - \alpha'\beta), & x\phi\alpha\alpha'(\gamma\delta' - \gamma'\delta) = w\delta\delta'(\alpha\beta' - \alpha'\beta), \\ y\phi\alpha\alpha'(\beta\gamma' - \beta'\gamma) = x\gamma\gamma'(\alpha\delta' - \alpha'\delta), & x\phi\delta\delta'(\alpha\beta' - \alpha'\beta) = w\alpha\alpha'(\gamma\delta' - \gamma'\delta). \end{array}$$

It may be added that: in plane $x = 0$,

$$\begin{array}{l} \text{intersection of } 1' \text{ lies on line } z : w = (\gamma\beta' - \alpha'\beta)\gamma\gamma' : \alpha\gamma\beta'\delta' - \alpha'\gamma'\beta\delta, \\ \text{intersection of } 2 \text{ lies on line } y : z = \alpha\gamma\beta'\delta' - \alpha'\gamma'\beta\delta : (\alpha\delta' - \alpha'\delta)\gamma\gamma', \end{array}$$

and that the line joining these intersections is the line 12. In plane $y = 0$,

$$\begin{array}{l} \text{intersection of } 2 \text{ lies on line } x : w = \alpha\gamma\beta'\delta' - \alpha'\gamma'\beta\delta : -(\beta\gamma' - \beta'\gamma)\delta\delta', \\ \text{intersection of } 3' \text{ lies on line } z : w = \alpha\gamma\beta'\delta' - \alpha'\gamma'\beta\delta : (\beta\delta' - \delta\beta')\delta\delta', \end{array}$$

and that the line joining these intersections is the line 23.

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In plane $z = 0$,

$$\begin{array}{l} \text{intersection of } 3' \text{ lies on line } x : y = (\gamma\delta' - \gamma'\delta)\alpha\alpha' : \alpha\gamma\beta'\delta' - \alpha'\gamma'\beta\delta, \\ \text{intersection of } 4 \text{ lies on line } x : w = -(\beta\gamma' - \beta'\gamma)\alpha\alpha' : \alpha\gamma\beta'\delta' - \alpha'\gamma'\beta\delta, \end{array}$$

and that the line joining these intersections is the line 34. And in plane $w = 0$,

$$\begin{array}{l} \text{intersection of } 4 \text{ is on line } y : z = (\alpha\delta' - \alpha'\delta)\beta\beta' : \alpha\gamma\beta'\delta' - \alpha'\gamma'\beta\delta, \\ \text{intersection of } 1' \text{ is on line } x : y = \alpha\gamma\beta'\delta' - \alpha'\gamma'\beta\delta : (\gamma\delta' - \gamma'\delta)\beta\beta', \end{array}$$

and that the line joining these intersections is the line 14.

The equations of the remaining ten lines of the surface may be obtained without difficulty, and also the forty-five triple planes, but I do not stop to effect this; the planes $x = 0$, $y = 0$, $z = 0$, $w = 0$, are, it is clear, triple planes, containing the lines 1, 2', 12; 2', 3, 23; 3, 4', 34; and 4', 1, 41 respectively.

If, to fix the ideas, the planes $x = 0, y = 0, z = 0, w = 0$ are taken to be those of the tetrahedron $ABCD$ ($x = BCD$, &c., as usual), then the edges AB, BC, CD, DA (but not the remaining opposite edges AC, BD) will be lines on the surface. Each plane of the tetrahedron, for instance ABC ($w = 0$), is met by the ten lines not contained therein in two vertices A, C , three points on the edge BA , three points on the edge BC , and two other points, viz. these are the intersections of the plane ABC by the lines 4 and 1'. For the construction of a model it is sufficient to determine the three points on each edge, and the two points, say in the plane ABC and in the plane DBC ($x = 0$) respectively; for then each of the remaining eight lines will be determined as a line joining two points in these two planes respectively. If in the first instance k is considered as a variable parameter, then the two points in the plane $w = 0$ are given as the intersections of two fixed lines by a variable line (14) rotating round the fixed point

$$\frac{x}{\alpha'} - \frac{y}{\beta'} + \frac{z}{\gamma'} = 0, \quad \frac{x}{\alpha} - \frac{y}{\beta} + \frac{z}{\gamma} = 0;$$

and the like as regards the two points in the plane $x = 0$. By making (with assumed values of the other parameters) the proper drawings for the two planes $w = 0, x = 0$, it is easy to fix upon a convenient value of the parameter k ; and I have in this manner succeeded in making a string model of the double-sixer; viz. the coordinates x, y, z, w are taken to be as the perpendicular distances of the current point from the faces of a regular tetrahedron (the coordinates being positive for an interior point); the values of $\alpha, \beta, \gamma, \delta$ were put $= 3, 4, 5, 6$ and those of $\alpha', \beta', \gamma', \delta' = 1, 1, 1, 1$; the value of k fixed upon as above was $k = -\frac{1}{2}$; this however brings the lines 2 and 4 too close together, and also their apparent intersection too close to their intersections with the line 6'; and it is probable that a slightly different value of k would be better.

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The results just obtained may be exhibited in a compendious form as follows:

1 is line BC :	$x = 0, z = 0;$	2' is line CD :	$x = 0, y = 0,$
3 is line DA :	$y = 0, z = 0;$	4' is line AB :	$z = 0, w = 0,$
5 meets CD :	$(z : w) = (\gamma : \delta),$	5 meets AB :	$(x : y) = (\alpha : \beta),$
6 meets CD :	$(z : w) = (\gamma' : \delta'),$	6 meets AB :	$(x : y) = (\alpha' : \beta'),$
6' meets BC :	$(y : z) = (\beta : \gamma),$	6' meets AD :	$(x : w) = (\alpha : \delta),$
5' meets BC :	$(y : z) = (\beta' : \gamma'),$	5' meets AD :	$(x : w) = (\alpha' : \delta'),$
1' meets AD :	$x : w = (\alpha' - k\alpha) : (\delta' - k\delta),$	1' meets BCD :	$\frac{-(\alpha' - k\alpha)(\gamma\delta' - \gamma'\delta)\beta\beta'}{(\delta' - k\delta)(\alpha\beta' - \alpha'\beta)\gamma\gamma'},$
1' meets ABC :	$\frac{(\alpha' - k\alpha)(\alpha\gamma\beta'\delta' - \alpha'\gamma'\beta\delta)}{(\alpha' - k\alpha)(\gamma\delta' - \gamma'\delta)\beta\beta'},$		
3' meets BC :	$y : z = (\beta' - k\beta) : (\gamma' - k\gamma),$	3' meets ACD :	$\frac{-(\beta' - k\beta)(\gamma\delta' - \gamma'\delta)\alpha\alpha'}{(\gamma' - k\gamma)(\alpha\beta' - \alpha'\beta)\delta\delta'},$
3' meets ABD :	$\frac{(\beta' - k\beta)(\alpha\gamma\beta'\delta' - \alpha'\gamma'\beta\delta)}{(\gamma' - k\gamma)(\alpha\beta' - \alpha'\beta)\delta\delta'},$		
2 meets AB :	$x : y = (\alpha' - k\alpha) : (\beta' - k\beta),$	2 meets BCD :	$\frac{(\alpha' - k\alpha)(\beta\gamma' - \beta'\gamma)\delta\delta'}{\delta' - k\delta},$
4 meets CD :	$z : w = (\gamma' - k\gamma) : (\delta' - k\delta),$	4 meets ABD :	$\frac{-(\delta' - k\delta)(\beta\gamma' - \beta'\gamma)\alpha\alpha'}{(\gamma' - k\gamma)(\alpha\delta' - \alpha'\delta)\beta\beta'}.$

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The four joining lines obtained from these intersections are

$$\begin{aligned} 1' \text{ and } 2 \text{ meet } BCD \text{ on line } & \left(-\frac{y}{\beta'} + \frac{z}{\gamma'} - \frac{w}{\delta'}\right) \frac{1}{\beta\delta} - k \left(-\frac{y}{\beta} + \frac{z}{\gamma} - \frac{w}{\delta}\right) \frac{1}{\beta'\delta'} = 0, \\ 2 \text{ and } 3' \text{ meet } CDA \text{ on line } & \left(\frac{x}{\alpha} + \frac{z}{\gamma} - \frac{w}{\delta}\right) \frac{1}{\alpha\gamma} - k \left(\frac{x}{\alpha'} + \frac{z}{\gamma'} - \frac{w}{\delta'}\right) \frac{1}{\alpha'\gamma'} = 0, \\ 3' \text{ and } 4 \text{ meet } DAB \text{ on line } & \left(\frac{x}{\alpha'} - \frac{y}{\beta'} - \frac{w}{\delta'}\right) \frac{1}{\beta\delta} - k \left(\frac{x}{\alpha} - \frac{y}{\beta} - \frac{w}{\delta}\right) \frac{1}{\beta'\delta'} = 0, \\ 4 \text{ and } 1' \text{ meet } ABC \text{ on line } & \left(\frac{x}{\alpha'} - \frac{y}{\beta'} + \frac{z}{\gamma'}\right) \frac{1}{\alpha\gamma} - k \left(\frac{x}{\alpha} - \frac{y}{\beta} + \frac{z}{\gamma}\right) \frac{1}{\alpha'\gamma'} = 0. \end{aligned}$$

For calculating the numerical values from the foregoing assumed data, say

1 : BC ; 2' : CD ; 3 : DA ; 4' : AB ,	
5 meets CD : $z = 45.5, w = 54.5$;	5 meets AB : $x = 42.9, y = 57.1$,
6 meets CD : $z = 50, w = 50$;	6 meets AB : $x = 50, y = 50$,
6' meets BC : $y = 44.4, z = 55.6$;	6' meets AD : $x = 33.3, w = 66.7$,
5' meets BC : $y = 50, z = 50$;	5' meets AD : $x = 50, w = 50$,
1' meets AD : $x = 44, z = 56$;	1' meets BCD : $y = -25.1, z = 39.9, w = 71.8$,
3' meets BC : $y = 48, z = 52$;	3' meets ABD : $x = 47.2, y = 141.7, w = -102.3$,
2 meets AB : $x = 47.8, y = 52.2$;	2 meets BCD : $y = 26.4, z = 44, w = 16.2$,
4 meets CD : $z = 48.1, w = 51.9$;	4 meets ABD : $x = 50.5, y = 187.6, w = -151.5$.

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$$\begin{aligned}
 1' \text{ and } 2 \text{ meet } BCD \text{ on line } & 35y - 32z + 30w = 0, \\
 2 \text{ and } 3' \text{ meet } CDA \text{ on line } & 26x + 22z - 21w = 0, \\
 3' \text{ and } 4 \text{ meet } DAB \text{ on line } & 8x - 7y - 6w = 0, \\
 4' \text{ and } 1 \text{ meet } ABC \text{ on line } & 52x - 47y - 44z = 0,
 \end{aligned}$$

which last four equations serve as a verification.

The outside numerical values are given in the manner most convenient for the construction of a drawing; viz. when the coordinates refer to a point on an edge of the tetrahedron, or say on the side of an equilateral triangle, then taking the length of this edge (or side) to be = 100, the numerical values are fixed so that the sum of the two coordinates may be = 100, and the two coordinates thus denote the distances from the extremities of the edge or side: but when the three coordinates belong to a point in the face of the tetrahedron, or say in the plane of an equilateral triangle, then the sum of the coordinates is made = 86.6, and the three coordinates thus denote the perpendicular distances from the sides of the triangle.

III.

It is possible to find on a cubic curve a double-sixer of points 1, 2, 3, 4, 5, 6 and 1', 2', 3', 4', 5', 6' such that any six points such as 1, 2, 3, 4', 5', 6' lie in a conic. In fact considering a cubic surface having upon it the double-sixer of lines 1, 2, 3, 4, 5, 6 and 1', 2', 3', 4', 5', 6', the section by any plane is a cubic curve meeting the lines, say in the points 1, 2, 3, 4, 5, 6, 1', 2', 3', 4', 5', 6': each of the lines 1, 2, 3 meets each of the lines 4', 5', 6', and consequently the six lines lie in a quadric surface: therefore the points 1, 2, 3, 4', 5', 6' lie in a conic: and so in the other cases; the number of the conics is of course = 60.

The cubic curve may be a given curve, and six of the points upon it (not being points on a conic) may also be taken to be given; for instance the points 1, 2, 3, 1', 4', 5'. For take through the points 2, 3 respectively any two lines 2, 3; through 1', 4', 5' respectively the lines 1', 4', 5' each meeting each of the lines 2, 3; and through 1 a line meeting each of the lines 4', 5'. It is easy to see that a cubic surface may be drawn through the cubic curve and the lines 1, 2, 3, 1', 4', 5'. For the passage through the cubic curve requires 9 conditions; the surface then passes through the point 2 and to make it pass through the line 2 requires 3 conditions; similarly the surface passes through the point 3, and to make it pass through the line 3 requires 3 conditions. The surface now passes through 1' and through the points of intersection of the line 1' with the lines 2, 3: to make it pass through the line 1' requires 1 condition; similarly to make it pass through the lines 4', 5', 1 requires in each case 1 condition; or there are in all 19 conditions, so that the cubic surface is completely determined. Take now through the points 1, 2, 3, 4', 5', a conic meeting the cubic in the point 6': then through the lines 1, 2, 3, 4', 5' we have a quadric surface passing through this

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conic, and therefore through 6': hence through 6' we may draw a line 6' meeting each of the lines 1, 2, 3; and since the cubic surface passes through the point 6' and also through the intersections of the line 6' with the lines 1, 2, 3, it passes through the line 6'. We complete in this manner by constructions in the plane of the cubic the system of the twelve points, viz. each new point is given as the intersection of the cubic curve by a conic drawn through five points of the cubic curve. It is

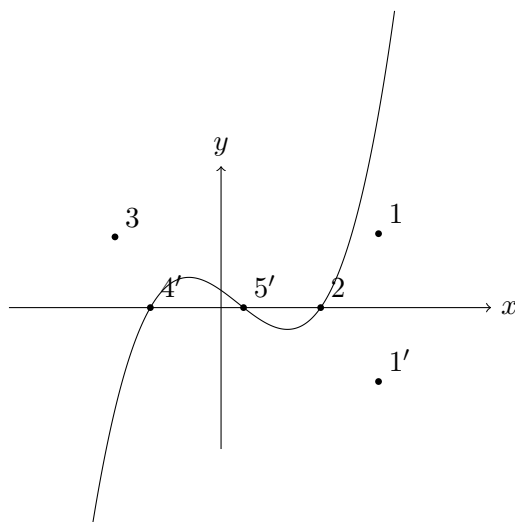
then shown as for the point $6'$ and the line $6'$ through it, that through each new point there can be drawn a line denoted by the same number and meeting each of the lines which it ought to meet, and hence lying on the cubic surface: the twelve points are thus the intersections of the plane of the cubic curve by the twelve lines of the double-sixer; and it follows that the six points which ought to lie in a conic (in every case where such conic has not been used in the plane construction) do actually lie in a conic.

I was anxious to construct such a double-sixer of points on a cubic curve; for this purpose I take the equation of the curve to be

$$y^2 = \left(1 - \frac{x}{a}\right) \left(1 - \frac{x}{b}\right) \left(1 - \frac{x}{c}\right),$$

or say for shortness $y^2 = X$; where, to fix the ideas, a, b are supposed to be positive, a greater than b ; and c to be negative.

The cubic curve is thus a parabola symmetrical in regard to the axis of x , and consisting of a loop and infinite branch; and I take upon it the points $1, 2, 3, 1', 4', 5'$ as shown in the figure,



viz. the coordinates of these points are as stated in the Table, where m is the x coordinate, and

$$\sqrt{M} = \sqrt{\left(1 - \frac{x}{a}\right) \left(1 - \frac{x}{b}\right) \left(1 - \frac{x}{c}\right)}$$

and so in other cases, $\sqrt{14} = 3.74165$.

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point	symbolic		numerical	
	x	y	x	y
1	m	$\sqrt{M}$	6	$\sqrt{14} = 3.742$
2	0	1	0	1
3	0	-1	0	-1
4	θ	$\sqrt{\Theta}$	$2 - \frac{75}{1369} = 1.945$	$-\frac{2280}{373}\sqrt{14} = -.168$
5	ϕ	$\sqrt{\Phi}$	$-1 + \frac{75}{361} = -.792$	$\frac{1110}{193}\sqrt{14} = .606$
6	m_1	$-\sqrt{M_1}$	$\frac{13}{3} = 4.333$	$-\frac{4}{9}\sqrt{14} = -1.641$
1'	m	$-\sqrt{M}$	6	$-\sqrt{14} = -3.742$
2'	σ	$\sqrt{\Sigma}$	$-\frac{1560(14 - \sqrt{14})}{(31\sqrt{14} - 5)^2} = 1.299$	.676
3'	τ	$\sqrt{T}$	$-\frac{1560(14 + \sqrt{14})}{(31\sqrt{14} + 5)^2} = 1.887$	.247
4'	c	0	-1	0
5'	b	0	2	0
6'	m_1	$\sqrt{M_1}$	$\frac{13}{3} = 4.333$	$\frac{4}{9}\sqrt{14} = 1.641$

The numerical values belong to the curve

$$y^2 = \left(1 - \frac{x}{3}\right) \left(1 - \frac{x}{2}\right) (1 + x)$$

and to $m = 6$.

Starting with the points 1, 2, 3, 1', 4', 5' we have to find the remaining points 6', 6, 4, 5, 2', 3'.

Point 6' by means of the conic 1234'5'6', as follows. The equation of the conic is

$$(x - b)(x - c) - bcy^2 + kxy = 0, \quad (2, 3, 4', 5'),$$

and making this pass through the point 1 ($x = m$, $y = \sqrt{M}$) we find

$$(m - b)(m - c) + ka\sqrt{M} = 0. \quad (1)$$

Hence taking the coordinates of 6' to be $m_1, \sqrt{M_1}$, we have

$$(m_1 - b)(m_1 - c) + ka\sqrt{M_1} = 0, \quad (6')$$

and thence

$$\frac{\sqrt{M_1}}{\sqrt{M}} = \frac{(m_1 - b)(m_1 - c)}{(m - b)(m - c)} = \frac{M_1(m - a)}{M(m_1 - a)}.$$

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that is,

$$\frac{\sqrt{M_1}}{\sqrt{M}} = \frac{(m_1 - b)(m_1 - c)}{(m - b)(m - c)} = \frac{m_1 - a}{m - a}.$$

We have thus for m_1 a quadric equation satisfied by $m = m_1$, so that throwing out the factor $m - m_1$, the equation is a linear one, viz. we find

$$m_1 = \frac{ma - ab - ac + bc}{m - a},$$

or, what is the same thing,

$$m_1 - a = \frac{(a - b)(a - c)}{m - a},$$

and thence also

$$\sqrt{M_1} = \frac{(a-b)(a-c)}{(m-a)^2} \sqrt{M},$$

viz. $\sqrt{M_1}$ is determined rationally in terms of $m, \sqrt{M}$; this is of course as it should be, since the point $6'$ is uniquely determinate.

Point 6 by means of the conic $2361'4'5'$. In precisely the same manner the coordinates are $m_1, -\sqrt{M_1}$, where $m_1, \sqrt{M_1}$ denote the same quantities as before.

Point 4 by means of the conic $2341'5'6'$. The equation of the conic is

$$Fx + Gy + H = \frac{1-y^2}{x}, \quad (2, 3),$$

where

$$\begin{aligned} Fb + H &= \frac{1}{b}, & (5'), \\ Fm_1 + G\sqrt{M_1} + H &= \frac{1-M_1}{m_1}, & (6'), \\ Fm - G\sqrt{M} + H &= \frac{1-M}{m}, & (1'), \end{aligned}$$

which give without difficulty

$$abcF = -a - c + P, \quad \sqrt{M} abcG = (m-b)(-m+P), \quad abcH = ab + ac + bc - bP,$$

where

$$P = 2a - c - \frac{2(a-c)(b-c)}{m+m_1-2c},$$

a quantity which will presently be expressed in terms of m only. And then

$$F\theta + G\sqrt{\Theta} + H = \frac{1-\Theta}{\theta},$$

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or say

$$F(\theta-b) + G\sqrt{\Theta} = \frac{1-\Theta}{\theta} - \frac{1}{b} = -(\theta-b) \left(\frac{1}{ab} + \frac{1}{bc} - \frac{\theta}{abc} \right),$$

that is,

$$(\theta-b)(abcF + a + c - \theta) + Gab\sqrt{\Theta} = 0,$$

viz. that is,

$$(\theta-b)(P-\theta) + (m-b) \frac{\sqrt{\Theta}}{\sqrt{M}} (P-m) = 0.$$

or, rationalising and throwing out the factor $\theta-b$, this is

$$(\theta-b)(\theta-P)^2 - (m-b)(m-P)^2 \frac{(\theta-a)(\theta-b)}{(m-a)(m-b)} = 0,$$

which is a cubic equation satisfied by $\theta = m$ and $\theta = m_1$; so that throwing out the factors $\theta-m, \theta-m_1$ we have for θ a linear equation.

Putting for shortness

$$A = (m-a)^2 - (a-b)(a-c), \quad B = (m-b)^2 - (b-c)(b-a), \quad C = (m-c)^2 - (c-a)(c-b),$$

the value of θ may be expressed in the forms

$$\theta - a = \frac{B^2}{C^2}(c-a), \quad \theta - b = \frac{A^2}{C^2}(c-b), \quad \theta - c = \frac{4(m-a)(m-b)(m-c)(b-c)(a-c)}{C^2}.$$

We have moreover

$$P - c = \frac{2(a - c)(m - b)(m - c)}{C}, \quad P - m = -\frac{(m - c)A}{C},$$

equations which express P in terms of m only; also

$$\theta - P = -\frac{2(a - c)(m - b)(m - c)B}{C^2},$$

and then

$$\sqrt{\Theta} = -\sqrt{M} \frac{\theta - b}{m - b} \frac{P - \theta}{P - m},$$

whence

$$\sqrt{\Theta} = 2\sqrt{M} (b - c)(c - a) \frac{AB}{C^3},$$

so that $\theta, \sqrt{\Theta}$ are now determined.

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Point 5 by means of the conic 2351'4'6'. The conic is

$$Fx + Gy + H = \frac{1 - y^2}{x}, \quad (2, 3),$$

where

$$\begin{aligned} Fc + H &= \frac{1}{c}, & (4'), \\ Fm_1 + G\sqrt{M_1} + H &= \frac{1 - M_1}{m_1}, & (6'), \\ Fm - G\sqrt{M} + H &= \frac{1 - M}{m}, & (1'). \end{aligned}$$

Everything is the same as for the point 4 except that b, c are interchanged; hence writing Q instead of P , and using A, B, C to denote as before, we have

$$abcF = -a - b + Q, \quad \sqrt{M} abcG = (m - c)(-m + Q), \quad abcH = ab + ac + bc - cQ,$$

and

$$\begin{aligned} \phi - a &= \frac{C^2(b - a)}{B^2}, & \phi - b &= \frac{4(m - a)(m - b)(m - c)(c - b)(a - b)}{B^2}, & \phi - c &= \frac{A^2(b - c)}{B^2}, \\ Q - b &= \frac{2(m - b)(m - c)(a - b)}{B}, & Q - m &= -\frac{A(m - b)}{B}, & \phi - Q &= -\frac{2(m - b)(m - c)C(a - b)}{B^2}, \end{aligned}$$

and

$$\sqrt{\Phi} = 2\sqrt{M} (c - b)(b - a) \frac{AC}{B^3},$$

which determine $\phi, \sqrt{\Phi}$.

Point 3' by means of the conic 1263'4'5'. The conic is

$$Fx + Gy + H = \frac{(x - b)(x - c)}{y}, \quad (4', 5'),$$

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and we have

$$G + H = bc, \quad (2)$$

$$Fm + G\sqrt{M} + H = \frac{(m - b)(m - c)}{\sqrt{M}}, \quad (1)$$

$$Fm_1 - G\sqrt{M_1} + H = \frac{(m_1 - b)(m_1 - c)}{-\sqrt{M_1}}, \quad (6)$$

Eliminating F , we have

$$G(m_1\sqrt{M} + m\sqrt{M_1}) + H(m - m_1) = \frac{m_1(m - b)(m - c)}{\sqrt{M}} \frac{m(m_1 - b)(m_1 - c)}{\sqrt{M_1}},$$

which is easily reduced first to

$$G \frac{2mm_1 - a(m + m_1)}{(m - a)\sqrt{M}} H(m - m_1) = (m + m_1) \frac{(m - a)(m - b)}{\sqrt{M}},$$

and then to

$$G\{aA + 2m(a - b)(a - c)\} - H \frac{(m - a)A}{\sqrt{M}} abc\{-A + 2m(m - a)\} = 0;$$

and combining herewith $G + H = bc$, we have

$$H = \frac{2bcm\{a(m - a) + (a - b)(a - c)\}}{aA + 2m(a - b)(a - c) + \frac{(m - a)A}{\sqrt{M}}},$$

$$G = bc - H;$$

and we have then

$$F(m + m_1) + G(\sqrt{M} - \sqrt{M_1}) + 2H = 0,$$

that is,

$$F\{2m(m - a) - A\} + G \frac{A\sqrt{M}}{m - a} + 2H(m - a) = 0,$$

or, what is the same thing,

$$F\{2m(m - a) - A\} = -bc \frac{A\sqrt{M}}{m - a} - H \left\{ 2(m - a) - \frac{A\sqrt{M}}{m - a} \right\}.$$

We then have

$$Fx + H = y \left(-G + \frac{(x - b)(x - c)}{y^2} \right) = y \left(-G - \frac{abc}{x - a} \right) = -\frac{y(Ha + Gx)}{x - a},$$

that is,

$$(Fx + H)^2 = -\frac{(x - b)(x - c)}{abc(x - a)} (Ha + Gx)^2.$$

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or

$$abc(x - a)(Fx + H)^2 + (x - b)(x - c)(Gx + Ha)^2 = 0.$$

Developing and throwing out the factor x , this is

$$\begin{aligned} & G^2x^3 + \{2aGH - (b + c)G^2 + abcF^2\}x^2 \\ & + \{a^2H^2 - 2a(b + c)GH + bcG^2 + abc(2FH - aF^2)\}x \\ & + \{-(b + c)a^2H^2 + 2abcGH + abc(H^2 - 2aFH)\} = 0. \end{aligned}$$

This must be satisfied by $x = m$, $x = m_1$; hence the left hand must be $G^2(x - m)(x - m_1)(x - \sigma)$, or equating the constant terms we have

$$G^2mm_1\sigma = aH\{-2abcF + 2bcG + (bc - ab - ac)H\},$$

which gives σ ; and we then have

$$\sqrt{\Sigma} = -\frac{\sigma - a}{G\sigma + Ha}(F\sigma + H),$$

but I have not attempted the further reduction of these expressions.

The numerical values for the example are

$$3F = \frac{-140 + 62\sqrt{14}}{5 + 21\sqrt{14}}, \quad G = \frac{-10 + 62\sqrt{14}}{5 + 21\sqrt{14}}, \quad H = \frac{-104\sqrt{14}}{5 + \sqrt{14}},$$

whence σ as in the Table.

Point $2'$ by means of the conic $1362'4'5'$. The equation of the conic is

$$Fx + Gy + H = \frac{(x - b)(x - c)}{y}, \quad (4', 5'),$$

where

$$-G + H = -bc, \tag{3}$$

$$Fm + G\sqrt{M} + H = \frac{(m - b)(m - c)}{\sqrt{M}}, \tag{1}$$

$$Fm_1 - G\sqrt{M_1} + H = \frac{(m_1 - b)(m_1 - c)}{-\sqrt{M_1}}, \tag{6}$$

which are the same as for point $3'$, if only we reverse the signs of F, H and $\sqrt{M}, \sqrt{M_1}$.

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Hence the formulae are

$$H = -\frac{2bcm\{a(m - a) + (a - b)(a - c)\}}{aA + 2m(a - b)(a - c) - \frac{(m - a)A}{\sqrt{M}}},$$

$$G = bc + H,$$

$$F\{2m(m - a) - A\} = -bc\frac{A\sqrt{M}}{m - a} - H\left\{2(m - a) + \frac{A\sqrt{M}}{m - a}\right\},$$

$$G^2mm_1\tau = aH\{-2abcF - 2bcG + (bc - ab - ac)H\},$$

which gives τ ; and then

$$\sqrt{T} = \frac{\tau - a}{G\tau - Ha}(F\tau + H),$$

which are also unreduced.

The numerical values are

$$3F = \frac{140 + 62\sqrt{14}}{5 - 21\sqrt{14}}, \quad G = \frac{-10 - 62\sqrt{14}}{5 - 21\sqrt{14}}, \quad H = \frac{-104\sqrt{14}}{5 - 21\sqrt{14}},$$

whence τ as in the Table.

[Printed p. 385; source PDF p. 449.]

522. NOTE ON THE THEORY OF INVARIANTS.

[From the *Mathematische Annalen*, vol. III. (1871), pp. 268–271.]

If two binary quantics $(a, \dots)(x, y)^n$, $(a', \dots)(x', y')^n$ are linearly transformable the one into the other, and if for the first of them P, Q are any two invariants whatever of the same degree, and P', Q' are the like invariants for the second of them, then we have

$$P : Q = P' : Q',$$

(or, what is the same thing, the absolute invariants have the same values for the two functions respectively); and the entire system of these equations constitutes only a $(n - 3)$ fold relation between the two sets of coefficients. But the converse theorem, viz. that if the entire system of equations is satisfied, the two functions are linearly transformable the one into the other, is only true *sub modo*.

For instance, considering the two binary sextics

$$(0, 0, 0, d, e, f, g)(x, y)^6 \quad \text{and} \quad (0, 0, 0, d', e', f', g')(x', y')^6,$$

or, what is the same thing,

$$(20d, 5e, 2f, g)(x, y)^3y^3 \quad \text{and} \quad (20d', 5e', 2f', g')(x', y')^3y'^3,$$

the invariants of the two functions respectively are each and all of them = 0, and yet the two functions are not in general linearly transformable the one into the other.

For they can be transformable only by the substitution

$$x = \lambda x' + \mu y', \quad y = \rho y';$$

or, what is the same thing, only if the cubic functions are transformable by the substitution $x = x' + ay'$, $y = y'$; and forming for these the seminvariants $ac - b^2$ and $a^2d - 3abc + 2b^3$ for the cubic $(a, b, c, d)(x, y)^3$, we have as the necessary condition for the transformability

$$(8df - 5e^2)^3 : (8d^2g - 12def + 5e^3)^2 = (8d'f' - 5e'^2)^3 : (8d'^2g' - 12d'e'f' + 5e'^3)^2.$$

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To deduce this result from the theory of the sextic function, I observe that denoting by A, B, C, Δ , the values of the quadrinvariant, the sextinvariant, and the discriminant, as given in Salmon's *Higher Algebra*, Ed. 2, pp. 202–211, then in the particular case $a = 0, b = 0$, we have

$$\begin{array}{llll} A = & -10d^2 & B = & d^4 \\ & +15ce & & -3cd^2e \\ & & & +c^2e^2 \end{array} \quad \begin{array}{ll} C = & -8d^6 \\ & +36cd^4e \\ & -39c^2d^2e^2 \\ & -8c^3e^3. \end{array} \quad \Delta = 0,$$

and hence forming the new invariants

$$B = 100B - A^2, \quad \Gamma = 1000C - 1200AB + 4A^3,$$

the values of these in the same particular case $a = b = 0$ are

$$B = 25c^2(8df - 5e^2) - 100c^3g,$$

$$\Gamma = -2500c^3(8d^2g - 12def + 5e^3) + 3000c^4(10eg - 9f^2).$$

Taking now A, B, C, Δ as the invariants of the sextic, one of the conditions for the transformation is $B^3 : C^2 = B'^3 : C'^2$.

In the particular case $a = b = c = 0$ and $a' = b' = c' = 0$, the invariants vanish and the equation is satisfied identically. But if we assume in the first instance only $a = b = 0$, $a' = b' = 0$, then the

terms contain the common factors c^6 and c'^6 respectively; and throwing these out, and then writing $c = 0, c' = 0$, we obtain the condition previously found in a different manner.

It will be observed that the condition is of the original form $P : Q = P' : Q'$, but with the difference that P, Q and the corresponding functions P', Q' , are not invariants. As possessing the foregoing property these functions may however be called “imperfect invariants,” it being understood that an imperfect invariant is not an invariant, and is not in any case included in the term “invariant” used without qualification.

And we may now establish the general theory as follows: Consider the similarly constituted special forms $(a, \dots)(x, y, z, \dots)^n$ and $(a', \dots)(x', y', z', \dots)^n$: to fix the ideas the coefficients $(a, \dots)$ may be regarded as homogeneous functions of the elements $(\alpha, \beta, \dots)$ which are either independent, or homogeneously connected together in any manner; and then the coefficients $(a', \dots)$ will be the like functions of the elements $(\alpha', \beta', \dots)$ which are either independent or (as the case may be) homogeneously connected in the like manner.

The entire series of functions $P, Q, \dots$ of $(\alpha, \beta, \dots)$, which are such that P, Q being of the same degree, and P', Q' being the like functions of $(\alpha', \beta', \dots)$, we have for the linearly transformable functions $(a, \dots)(x, y, z, \dots)^n$ and $(a', \dots)(x', y', z', \dots)^n$ the relation

$$P : Q = P' : Q',$$

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may be called the “perfect and imperfect invariants” of $(a, \dots)(x, y, z, \dots)^n$; and the relation in question be briefly referred to by the expression that the perfect and imperfect invariants are proportional.

We have then the theorem that if the two functions $(a, \dots)(x, y, z, \dots)^n$ and $(a', \dots)(x', y', z', \dots)^n$ are linearly transformable the one into the other, the two functions have their perfect and imperfect invariants proportional; and *conversely* the theorem, that two functions which have their perfect and imperfect invariants proportional, are linearly transformable the one into the other.

There is thus a wide field of inquiry in regard to the imperfect invariants, even of a binary function, but still more so as to those of a ternary or quaternary function representing a curve or surface possessed of singularities.

We have in what precedes the explanation of an error into which I fell in my paper “On the transformation of plane curves,” *Proc. Lond. Math. Soc.*, vol. I. No. 3, Oct. 1865, [384], see Arts. Nos. 27–30. Considering a given curve of deficiency D and, by means of a system of $D - 3$ points chosen at pleasure on the curve, transforming this into a curve of the order $D + 1$ with deficiency D ; then for any two of the transformed curves (that is, two curves obtained by means of different systems of the $D - 3$ points) I showed that these had the same absolute invariants—or in the language of the present paper, that they had their invariants proportional, and I thence inferred that the two transformed curves were linearly transformable the one into the other—whereas, to sustain this conclusion, it is necessary that the two curves should have their perfect and imperfect invariants proportional; and this was in no wise proved. That the two transformed curves are not in fact linearly transformable the one into the other has since been shown *a posteriori* by Dr Brill in the particular case $D = 4$. Riemann’s conclusions, with which my own were at variance, are thus correct.

I remark that if a binary function of an odd or even degree $n = 2p + 1$ or $= 2p$, has $p + 1$ equal factors, then the invariants all of them vanish; but the equality of the $p + 1$ factors implies only a p -fold relation between the coefficients; that is, the vanishing of all the invariants gives only a p -fold relation between the coefficients, viz. the relation is $\frac{1}{2}(n - 1)$ fold or $\frac{1}{2}n$ -fold according as n is odd or even. Thus for a sextic function the equations $A = 0, B = 0, C = 0, \Delta = 0$ constitute only a 3-fold relation between the coefficients.

Similarly if the function has p equal factors, then every invariant is a mere numerical multiple of a power of one and the same function Θ ; so that the vanishing invariants can be at once formed.

And we have thus only a $(p - 1)$ fold relation between the coefficients, viz. the relation is $\frac{1}{2}(n - 3)$ fold or $\frac{1}{2}(n - 2)$ fold according as n is odd or even.

Cambridge, 4 August, 1870.

[Printed p. 388; source PDF p. 452.]

523. ON THE TRANSFORMATION OF UNICURSAL SURFACES.

[From the *Mathematische Annalen*, vol. III. (1871), pp. 469–474.]

I consider the question of the transformation (*Abbildung auf einer Ebene*) of unicursal surfaces. Taking (x, y, z, w) for the coordinates of a point on the surface, (x', y', z') for those of the corresponding point on the plane; then if X', Y', Z', W' denote each of them a function $(x', y', z')^{n'}$, the equations of transformation are

$$x : y : z : w = X' : Y' : Z' : W' :$$

and assuming that each of the curves

$$X' = 0, \quad Y' = 0, \quad Z' = 0, \quad W' = 0$$

(or, what is the same thing, the general curve

$$aX' + bY' + cZ' + dW' = 0)$$

passes once through each of α_1 points, twice through each of α_2 points, $\dots$, r times through each of α_r points (for convenience I write α_r instead of α'_r); and writing also

$$n = n'^2 - \sum r^2 \alpha_r,$$

$$0 = \frac{1}{2}n'(n' + 3) - 3 - \Theta - \frac{1}{2} \sum r(r + 1) \alpha_r,$$

(where Θ is = 0 or positive except in the case of special relations between the positions of the fixed points $\alpha_1, \alpha_2, \dots, \alpha_r$), which equations give

$$-n = 3n' - 6 - 2\Theta - \sum r \alpha_r;$$

then the order of the surface is $= n$, and the order of the nodal curve is

$$b = \frac{1}{2}(n - 2)(n - 3) + \Theta.$$

I assume that the nodal curve has h apparent double points and t actual triple points, but no stationary points, so that q being the class, we have

$$q = b^2 - b - 2h - 6t;$$

and I endeavour to find these numbers q, t, h .

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For this purpose, imagine through the nodal curve a surface of the order k , which therefore meets the surface besides in a curve of the order $nk - 2b$; this curve I call the k -thic residue of the nodal curve, or simply the “residue.” The projection (*Abbildung*) of the complete intersection kn is a curve of the order kn' passing kr times through each of the points α_r : this is made up of the projection of the nodal curve once, and of the projection of the residue. But as shown by Dr Clebsch

the projection of the nodal curve is of the order $(n-4)n' + 3$, and it passes $(n-4)r + 1$ times through each of the points α_r ; hence the projection of the residue is of the order $(k-n+4)n' - 3$, and it passes $(k-n+4)r - 1$ times through each of the points α_r . I assume that the projection of the residue is the general curve which satisfies the foregoing conditions, viz. that the residue, and its projection as defined by the foregoing conditions, depend each of them on the same number of constants. The necessity for this I confess by no means obvious: but take as an illustration Steiner's quartic surface as transformed by the equations

$$x : y : z : w = x'^2 : y'^2 : z'^2 : (x' + y' + z')^2 :$$

the nodal curve consists of three lines meeting in a point, the quadric residue is the remaining intersection of the surface by a quadric cone passing through the three lines; and the projection thereof is a line; the quadric cone, and therefore the conic, each depend upon 2 constants; and the line which is the projection of the conic depends upon the same number (2) of constants: at all events I make the assumption provisionally.

Now in the projection of the residue, we have twice the number of constants

$$= [(k-n+4)n' - 3](k-n+4)n' - 2 \sum [(k-n+4)r - 1](k-n+4)r \alpha_r,$$

viz. this is

$$= (k-n+4)^2 (n'^2 - \sum r^2 \alpha_r) + (k-n+4) (-3n' + 2 \sum r \alpha_r),$$

or, what is the same thing, it is

$$= (k-n+4)^2 n + (k-n+4)(n-6-2\Theta),$$

viz. reducing, and replacing Θ by its value $-\frac{1}{2}(n-2)(n-3) + b$, the number in question is

$$= k^2 n + k(-n^2 + 4n - 2b) + 2(n-4)b.$$

Now k being $=$ or $> n-3$, the curve of intersection of a given surface n by a surface k depends on

$$\frac{1}{6}(k+1)(k+2)(k+3) - \frac{1}{6}(k-n+1)(k-n+2)(k-n+3) - 1$$

constants; and making the surface k to pass through the curve b we have to subtract herefrom $(k+1)b - \frac{1}{2}q - 2t$; that is, for the residue, twice the number of constants is

$$\begin{aligned} &= \frac{1}{3}(k+1)(k+2)(k+3) - \frac{1}{3}(k-n+1)(k-n+2)(k-n+3) \\ &\quad - 2 - 2(k+1)b + q + 4t, \end{aligned}$$

viz. this is

$$= k^2 n + k(-n^2 + 4n - 2b) + \frac{1}{3}(n-1)(n-2)(n-3) - 2b + q + 4t.$$

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Hence comparing the two expressions in question we have

$$2(n-4)b = \frac{1}{3}(n-1)(n-2)(n-3) - 2b + q + 4t,$$

that is,

$$0 = \frac{1}{3}(n-1)(n-2)(n-3) - 2(n-3)b + q + 4t,$$

or, as I prefer to write it,

$$q + 4t = 2(n-3)b - \frac{1}{3}(n-1)(n-2)(n-3),$$

which agrees with a more general formula in my “Memoir on the theory of Reciprocal Surfaces,” *Phil. Trans.* vol. CLIX. (1869), [411], see p. 227, [Coll. Math. Papers, vol. VI. p. 356]. I consider any two residues, a k -thic residue and a l -thic residue; to each intersection of these there corresponds an intersection of their projections: or the number of intersections of the two residues must be equal to that of the two projections. Now the projections being (as above)

$$\begin{aligned} &\text{order } (k - n + 4)n' - 3 \text{ passing } (k - n + 4)r - 1 \text{ times through each point } \alpha_r, \\ &\text{order } (l - n + 4)n' - 3 \text{ passing } (l - n + 4)r - 1 \text{ times through each point } \alpha_r, \end{aligned}$$

the number of the intersections in question is

$$\begin{aligned} &= [(k - n + 4)n' - 3][(l - n + 4)n' - 3] \\ &\quad - \sum [(k - n + 4)r - 1][(l - n + 4)r - 1]\alpha_r + \omega, \end{aligned}$$

where for a reason which will be afterwards explained I have added the term ω : this is

$$\begin{aligned} &= (k - n + 4)(l - n + 4) \left(n'^2 - \sum r^2 \alpha_r \right) \\ &\quad + (k + l - 2n + 8) \left(-3n' + \sum r \alpha_r \right) + 9 - \left(\sum \alpha_r - \omega \right), \end{aligned}$$

viz. it is

$$= (k - n + 4)(l - n + 4)n + (k + l - 2n + 8)(n - 6 - 2\Theta) + 9 - \left(\sum \alpha_r - \omega \right),$$

viz. substituting for Θ its value $-\frac{1}{2}(n - 2)(n - 3) + b$, and reducing, the number is

$$= kln - 2(k + l)b - n^3 + 8n^2 - 16n + 9 + 4(n - 4)b - \left(\sum \alpha_r - \omega \right).$$

But the surfaces n, k, l , having in common the curve b which is a nodal curve on n , besides intersect in

$$kln - b(n + 2k + 2l - 4) + 2q + 9t$$

points (Salmon’s *Geometry of Three Dimensions*, 2nd Ed. p. 283, except that in the formula as there given the singularity t is not taken account of); that is, the number of intersections of the two residues is

$$kln - b(n + 2k + 2l - 4) + 2q + 9t,$$

which is equal to the number of intersections of the two projections; or comparing the numbers in question we have

$$-n^3 + 8n^2 - 16n + 9 + 4(n - 4)b - \left(\sum \alpha_r - \omega \right) = -(n - 4)b + 2q + 9t,$$

that is,

$$2q + 9t = 5(n - 4)b - n^3 + 8n^2 - 16n + 9 - \left(\sum \alpha_r - \omega \right).$$

Two $(n + \lambda)$ -thic residues meet in $n(\lambda + 4)(\lambda + 6) - 12\lambda - 39 - 4(\lambda + 4)\Theta - \left(\sum \alpha_r - \omega \right)$ points; in particular, two $(n - 3)$ -thic residues meet in $3n - 3 - 4\Theta - \left(\sum \alpha_r - \omega \right)$ points; and two $(n - 2)$ -thic residues meet in $8n - 15 - 8\Theta - \left(\sum \alpha_r - \omega \right)$ points.

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But we have already found

$$2q + 8t = 4(n - 3)b - \frac{1}{2}n^3 + 4n^2 - \frac{25}{2}n + 4,$$

and we have therefore

$$t = (n - 8)b - \frac{1}{2}n^3 + 4n^2 - \frac{7}{2}n + 5 - \left(\sum \alpha_r - \omega \right),$$

and

$$q = -2(n-13)b + n^3 - 14n^2 + 31n - 18 + 4(\sum \alpha_r - \omega).$$

I obtain these results in a different manner by investigating expressions for the deficiency (*Geschlecht*) of the nodal residue $nk - 2b$ and for that of its projection.

First for the projection, we have

$$\begin{aligned} \text{Twice Deficiency} &= [(k-n+4)n' - 4][(k-n+4)n' - 5] \\ &\quad - \sum [(k-n+4)r - 1][(k-n+4)r - 2]\alpha_r + 2\omega, \end{aligned}$$

where I have added the term 2ω , as afterwards explained: this is

$$= (k-n+4)^2 \left(n'^2 - \sum r^2 \alpha_r \right) + (k-n+4) \left(-9n' + 3 \sum r \alpha_r \right) + 20 - 2(\sum \alpha_r - \omega),$$

viz. it is

$$= (k-n+4)^2 n + (k-n+4)(3n - 18 - 6\Theta) + 20 - 2(\sum \alpha_r - \omega),$$

or substituting for Θ its value $-\frac{1}{2}(n-2)(n-3) + b$ and reducing, it is

$$= k^2 n + k(n^2 - 4n - 6b) - 2n^3 + 16n^2 - 32n + 20 + 6(n-4)b - 2(\sum \alpha_r - \omega).$$

Next as regards the residue, the number h' of its apparent double points is obtained in terms of h and t by the formula

$$8h + 6t - 2h' = (kn - 4b)(k-1)(n-1) - 2b(k-1),$$

(Salmon, l.c., p. 284, except that the singularity t is not there taken account of); and we thence have

$$\begin{aligned} \text{Twice Deficiency} &= (kn-1)(kn-2) - 2h' \\ &= kn(k+n-4) + 4b^2 + b(-4n-6k+12) + 2 - 8h - 6t, \end{aligned}$$

or introducing q instead of h by the formula $4b^2 - 8h = 4q + 4b + 24t$, this is

$$= kn(k+n-4) + b(-4n-6k+12) + 4q + 4b + 2 + 18t,$$

viz. it is

$$= k^2 n + k(n^2 - 4n - 6b) - 4(n-4)b + 4q + 18t.$$

So that comparing with the deficiency of the projection we have

$$-2n^3 + 16n^2 - 32n + 20 + 6(n-4)b - 2(\sum \alpha_r - \omega) = -4(n-4)b + 4q + 2 + 18t,$$

that is,

$$2q + 9t = 5(n-4)b - n^3 + 8n^2 - 16n + 9 - (\sum \alpha_r - \omega),$$

the same result as before.

[523] ON THE TRANSFORMATION OF UNICURSAL SURFACES

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The necessity for the term ω appears by the consideration that if we apply to the plane figure a Cremona-transformation, thus obtaining a new transformation of the surface, the value of Σa_r will in general be altered; whereas the expressions for q, t should it is clear remain unaltered; and it arises as follows, viz. for certain transformations of the surface the curve of the order $(k - n + 4)n' - 3$, passing $(k - n + 4)r - 1$ times through each point a_r and assumed to be the projection of the residue, is not an indecomposable curve but contains a certain number ω of factors (each belonging to a unicursal curve definable by means of the number of its passages through the several points a_r), which factors are to be rejected in order to obtain the equation of the proper residue. Thus reverting to the transformation

$$x : y : z : w = x'^2 : y'^2 : z'^2 : (x' + y' + z')^2$$

of Steiner's surface, the projection of the quadric residue was (as already remarked) a line; applying to the plane figure the ordinary quadric (or inverse) transformation we introduce three fixed points, ($a_2 = 3$), say these are A, B, C ; viz. in the new transformation of the surface the projection of any plane section is a quartic curve having a node at each of the fixed points: the projection of the residue ought clearly to be a conic through the three points; but according to the general formula it is a quintic having at each of these points a triple point: the quintic is in fact made up of the lines BC, CA, AB and of the conic which is the proper residue; viz. in the case in question there are 3 factors thrown out, or we have $\omega = 3$. To apply this to the second investigation of $2q + 9t$, by comparison of the two deficiencies, observe that in general if a curve is made up of $\omega + 1$ indecomposable curves, the deficiency of the compound curve is equal to the sum of the deficiencies of the component curves $-\omega$; hence if ω of the curves are unicursal, the deficiency of the compound curve is equal to that of the remaining curve $-\omega$; or, what is the same thing, the deficiency of the remaining curve is equal to that of the compound curve $+\omega$; and the addition of the term $+\omega$ to the expression for the deficiency is thus accounted for. It is easy to see that a like explanation applies to the first investigation of $2q + 9t$.

I further remark, reverting to the equations

$$x : y : z : w = X' : Y' : Z' : W'$$

of the transformation, that the product of the ω factors is given as the common factor (if any) of the Jacobians

$$J(Y', Z', W'), \quad J(Z', W', X'), \quad J(W', X', Y') \quad \text{and} \quad J(X', Y', Z').$$

Such common factor exists whenever we can by a Cremona-transformation of the plane figure reduce the number of the points a_r upon which the transformation of the surface depends; viz. for any given transformation of the surface, ω is equal to the excess of Σa_r above the minimum value of Σa_r , or, what is the same thing, $\Sigma a_r - \omega$ is equal to the minimum value of Σa_r , and is thus independent of the particular transformation. And of course if Σa_r has this minimum value, viz. if the transformation is such that the number of the points a_r cannot be reduced by any Cremona-transformation of the plane figure, then we have $\omega = 0$.

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I presume that for the most simple transformation, that is, when n' has its least value, Σa_r has also its least value, and consequently that ω is $= 0$.

Recapitulating, the results obtained are

$$\begin{aligned} q &= -2(n - 13)b + n^3 - 14n^2 + 31n - 18 + 4(\Sigma a_r - \omega), \\ t &= (n - 8)b - \frac{1}{3}n^3 + 4n^2 - \frac{26}{3}n + 5 - (\Sigma a_r - \omega), \end{aligned}$$

where it will be recollected that

$$b = \frac{1}{2}(n-2)(n-3) + \Theta;$$

the formulae are verified in the several cases:

n'	a_1	a'_2	n	Θ	ω	b	q	t	
2	2	0	2	0	1	0	0	0	Quadric surface
2	1	0	3	1	0	1	0	0	Cubic scroll
2	0	0	4	2	0	3	0	1	Steiner's quartic surface
3	6	0	3	0	0	0	0	0	Cubic surface
3	5	0	4	1	0	2	2	0	Quartic with nodal conic
2	8	1	4	0	0	1	0	0	Do. with nodal line
2	7	1	5	-1	0	4	8	0	Quintic with nodal quadriquadric
2	11	0	5	0	0	3	4	0	Do. with nodal skew cubic
2	12	2	5	-1	0	2	0	0	Do. with two non-intersecting nodal lines

which are the transformations chiefly as yet examined: but the first-mentioned case (quadric surface, generalised stereographic projection), although as stated the formulae are verified with the value $\omega = 1$, does not really come under the foregoing theory. It is interesting to see that they are verified in the last-mentioned case, belonging to a negative value of Θ , that is, to a special system of fixed points.

Cambridge, Dec. 5, 1870.

[524] ON THE DEFICIENCY OF CERTAIN SURFACES

[From the *Mathematische Annalen*, vol. III. (1871), pp. 526–529.]

p. 394 [524; source PDF p. 458]

If a given point or curve is to be an ordinary or singular point or curve on a surface of the order n , this imposes on the surface a certain number of conditions, which number may be termed the “Postulation”; thus “Postulation of a given curve *qua* i -tuple curve on a surface n ” will denote the number of conditions to be satisfied by the surface in order that the given curve may be an i -tuple curve on the surface.

The “deficiency” (Flächengeschlecht) of a given surface of the order n is

$$= \frac{1}{6}(n-1)(n-2)(n-3)$$

less deficiency-value of the several singularities; viz. as shown by Dr Noether, if the surface has a given i -tuple curve, the deficiency-value hereof is

$$= \text{Postulation of the curve } qua (i-1) \text{ tuple curve on a surface } n-4;$$

and if the surface has an i -conical point, the deficiency-value hereof is

$$= \text{Postulation of the point } qua (i-2) \text{ conical point on a surface } n-4;$$

viz. this is $= \frac{1}{6}i(i-1)(i-2)$, and is thus independent of the order of the surface.

I remark that if the tangent-cone at the i -conical point has δ double lines and κ cuspidal lines, then the deficiency-value is

$$= \frac{1}{6}i(i-1)(i-2) + (i-2)(n-i-1)(\delta + \kappa).$$

In the case of a double or cuspidal curve $i = 2$, and the deficiency-value is

$$= \text{Postulation of given curve } qua \text{ simple curve on a surface } n-4;$$

and so for an ordinary conical point $i = 2$, and the deficiency-value is $= 0$: results which were first obtained by Dr Clebsch.

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I found in this manner the expression for the deficiency of a surface n having a double and cuspidal curve and the other singularities considered in my “Memoir on the Theory of Reciprocal Surfaces,” *Phil. Trans.* vol. CLIX. (1869), [411, *Coll. Math. Papers*, vol. VI. p. 356]; viz. this was

$$D = \frac{1}{6}(n-1)(n-2)(n-3) - (n-3)(b+c) + \frac{1}{2}(q+r) + 2t + \frac{7}{2}\beta + \frac{3}{2}\gamma + i - \frac{1}{8}\theta,$$

where we have

- b , order of double curve,
- q , class of Do.,
- c , order of cuspidal curve,
- r , class of Do.,
- β , number of intersections of the two curves, stationary points on b ,
- γ , number of intersections, stationary points on c ,
- i , number of intersections, not stationary points on either curve,
- θ , number of certain singular points on c , the nature of which I do not completely understand; it is here taken to be $= 0$.

Before going further I remark that

$$\begin{aligned} \text{Postulation of right line } qua \text{ } i\text{-tuple on surface } n &= \frac{1}{2}i(i+1)n - \frac{1}{6}i(i+1)(2i-5) \\ &= \frac{1}{6}i(i+1)(3n-2i+5). \end{aligned}$$

Whence if a surface n has an i -tuple right line, the deficiency-value hereof is

$$= \frac{1}{6}i(i-1)(3n-2i-5),$$

or we have

$$\begin{aligned} D &= \frac{1}{6}(n-1)(n-2)(n-3) - \frac{1}{6}i(i-1)(3n-2i-5) \\ &= \frac{1}{6}(i-n+1)(i-n+2)(2i+n-3); \end{aligned}$$

so that $D = 0$ if either $i = n-1$ or $i = n-2$; the former case is that of a scroll (skew surface) with an $(n-1)$ tuple right line, the latter that of a surface with an $(n-2)$ tuple line: whence (as shown by Dr Noether) such surface is rationally transformable into a plane.

For a surface of the order n with an i -conical point where the tangent cone has δ double lines and κ cuspidal lines, we have

$$\begin{aligned} D &= \frac{1}{6}(n-1)(n-2)(n-3) \\ &\quad - \left\{ \frac{1}{6}i(i-1)(i-2) + (i-2)(n-i-1)(\delta + \kappa) \right\} \\ &= \frac{1}{6}(n-i-1)\{n^2 + n(i-5) + i^2 - 4i + 6 - 6(i-2)(\delta + \kappa)\}; \end{aligned}$$

viz. for $i = n-1$ this is $D = 0$ (in fact, a surface n with an $(n-1)$ conical point is at once seen to be rationally transformable into a plane): and for $i = n$, that is, for a cone of the order n , we have

$$D = -\frac{1}{2}(n-1)(n-2) + (n-2)(\delta + \kappa) - (n-3)(\delta + \kappa),$$

where the last term $-(n-3)(\delta + \kappa)$ is added because in the present case the surface has the δ double lines and the κ cuspidal lines.

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The formula therefore gives

$$D = -\frac{1}{2}(n-1)(n-2) + \delta + \kappa;$$

viz. this is equal to the deficiency of the plane sections taken negatively.

I find that the same property exists first in the case of a scroll (skew surface) having only a double curve; and secondly in the case of a torse (developable surface) having a cuspidal curve with the ordinary singularities; and this being so there can I think be no doubt but that it is true for any scroll or torse whatever—viz. that for any ruled surface whatever the deficiency is equal to that of the plane section taken negatively.

First, for the scroll, we have

$$D = \frac{1}{6}(n-1)(n-2)(n-3) - (n-3)b + \frac{1}{2}q + 2t,$$

which should be

$$= -\frac{1}{2}(n-1)(n-2) + h.$$

Salmon's equations give in the case of a scroll

$$3t = (n-4)\{3b - n(n-2)\}, \quad q = n(n-2)(n-5) - 2(n-6)b,$$

and with these values the relation is at once verified.

Secondly, for the torse; changing the notation into that used for the singularities of the curve and torse, we have

$$D = \frac{1}{6}(n-1)(n-2)(n-3) - (n-3)x + \frac{1}{2}q + 2t + \frac{7}{2}\beta + \frac{3}{2}\gamma + i,$$

which should be

$$= -\frac{1}{2}(m-1)(m-2) + h + \beta.$$

We have $q = r(n-3) - 3x$, and substituting this value and expressing everything in terms of r, m, n by means of the formulae

$$\begin{aligned} x &= \frac{1}{2}(r^2 - r - n - 3m), & a &= \frac{1}{2}n - 3r + 3m, \\ \beta &= n - 3r + 3m, \\ t &= \frac{1}{2}\{r^3 - 3r^2 - 58r - 3r(n+3m) + 42n + 78m\}, \\ \gamma &= rm + 12r - 14m - 6n, \end{aligned}$$

we have after all reductions

$$D = -\frac{1}{2}(m+n) + r - 1 = -\frac{1}{2}(m-1)(m-2) + h + \beta.$$

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We have thus a class of surfaces of negative deficiency; viz. any rational transformation of a cone for which the plane section has a given (positive) deficiency produces such a surface: and I think it may be assumed conversely that a surface of negative deficiency is always the rational transformation of a cone for which the deficiency is equal to that of the surface taken with the reverse sign. As an instance, take a quintic surface having a nodal conic and two 3-conical (cubiconical) points (this of course implies that the line joining the two cubiconical points is a line on the surface); the formula for the deficiency is ($n = 5, b = 2, q = 2, r = 0, t = 0$)

$$D = \frac{1}{6}(n-1)(n-2)(n-3) - (n-3)b + \frac{1}{2}q + 2t - 2,$$

(viz. a term -1 for each of the cubiconical points)

$$= 4 - 4 + 1 - 2, \quad = -1.$$

Such a surface can be obtained as the quadric inverse of a cubic cone; viz. taking for the vertex the point $x : y : z : w = \alpha : \beta : \gamma : \delta$ and the cone to pass through the point $x = 0, y = 0, z = 0$, the equation of the cone is

$$(\delta x - \alpha w, \delta y - \beta w, \delta z - \gamma w)^3 = 0,$$

where

$$(\alpha, \beta, \gamma)^3 = 0.$$

Taking Q a quadric function $(x, y, z)^2$, the transformation in question consists in the change of x, y, z, w into xw, yw, zw, Q ; viz. the new equation, rejecting the factor w which divides out, is

$$w^{-1}(\delta xw - \alpha Q, \delta yw - \beta Q, \delta zw - \gamma Q)^3 = 0,$$

which is a quintic surface, having the two cubiconical points $x = 0, y = 0, z = 0$ and

$$x : y : z : w = \alpha : \beta : \gamma : \delta^{-1}Q_0$$

(where Q_0 is the value of Q on writing therein α, β, γ in place of x, y, z): and having the nodal conic $w = 0, Q = 0$.

Cambridge, 5 Jan. 1871.

[525] AN EXAMPLE OF THE HIGHER TRANSFORMATION OF A BINARY FORM

[From the *Mathematische Annalen*, vol. IV. (1871), pp. 359–361.]

p. 398 [525; source PDF p. 462]

The quartic

$$(1) \quad (a, b, c, d, e)(x, y)^4$$

is by means of the two quadrics

$$(2) \quad (\alpha, \beta, \gamma)(x, y)^2 \quad \text{and} \quad (\alpha', \beta', \gamma')(x, y)^2$$

transformed into

$$(3) \quad (a_1, b_1, c_1, d_1, e_1)(x_1, y_1)^4,$$

that is, eliminating x, y from

$$(a, b, c, d, e)(x, y)^4 = 0, \quad (\alpha x_1 + \alpha' y_1, \beta x_1 + \beta' y_1, \gamma x_1 + \gamma' y_1)(x, y)^2 = 0,$$

we obtain

$$(a_1, b_1, c_1, d_1, e_1)(x_1, y_1)^4 = 0.$$

It is required to express the invariants of (3) in terms of the simultaneous invariants of (1) and (2).

Write

$$P, Q, R = \alpha x_1 + \alpha' y_1, \quad \beta x_1 + \beta' y_1, \quad \gamma x_1 + \gamma' y_1;$$

the equations from which (x, y) have to be eliminated are

$$(a, b, c, d, e)(x, y)^4 = 0, \quad (P, Q, R)(x, y)^2 = 0.$$

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and the result of the elimination therefore is

$$\begin{vmatrix} a & 4b & 6c & 4d & e \\ & a & 4b & 6c & 4d & e \\ & & P & 2Q & R \\ & & P & 2Q & R \\ P & 2Q & R & & \end{vmatrix} = 0,$$

viz. this determinant is the transformed quartic $(a_1, b_1, c_1, d_1, e_1)(x_1, y_1)^4$.

The developed expression of the determinant is

$$\begin{aligned} & a^2 R^4 - 8abQR^3 + \left(\begin{matrix} -12ac \\ +16b^2 \end{matrix} \right) PR^3 - 24acQ^2R^2 + \left(\begin{matrix} 24ad \\ -48bc \end{matrix} \right) PQR^2 - 32adQ^3R \\ & + \left(\begin{matrix} 2ae \\ -32bd \\ +36c^2 \end{matrix} \right) P^2R^2 + \left(\begin{matrix} -16ae \\ +64bd \end{matrix} \right) PQ^2R + 16aeQ^4 \\ & + \left(\begin{matrix} 24be \\ -48cd \end{matrix} \right) P^2QR - 32bePQ^3 + \left(\begin{matrix} -12ce \\ +16d^2 \end{matrix} \right) P^3R + 24ceP^2Q^2 \\ & - 8deP^3Q + e^2P^4. \end{aligned}$$

So that writing for P, Q, R their values, we have the transformed function $(a_1, b_1, c_1, d_1, e_1)(x_1, y_1)^4$, the coefficients being of the forms

$$\begin{aligned} a_1 &= (a, b, c, d, e)^2(\alpha, \beta, \gamma)^4, \\ b_1 &= (\alpha\beta)^2(\alpha, \beta, \gamma)^3(\alpha', \beta', \gamma'), \\ &\dots \\ e_1 &= (\alpha\gamma)^2(\alpha', \beta', \gamma')^4. \end{aligned}$$

Writing I, J for the invariants of the quartic (1), and

$$\begin{aligned} A &= 4(\alpha\beta' - \alpha'\beta)(\beta\gamma' - \beta'\gamma) - (\gamma\alpha' - \gamma'\alpha)^2, \\ B &= (e, c, a, b, c, d)(\alpha\beta' - \alpha'\beta, \gamma\alpha' - \gamma'\alpha, \beta\gamma' - \beta'\gamma)^2, \end{aligned}$$

we have I, J, A, B simultaneous invariants of the forms (1) and (2). Putting moreover $\nabla = I^3 - 27J^2$, and writing I_1, J_1, ∇_1 for the like invariants of the form (3), I find

$$\begin{aligned} I_1 &= 4 \left(4IB^2 + 12JAB + \frac{1}{3}I^2A^2 \right), \\ J_1 &= 8 \left\{ 8JB^3 + \frac{1}{3}I^2AB^2 + 2IJA^2B + \left(2J^2 - \frac{1}{27}I^3 \right) A^3 \right\}, \end{aligned}$$

and thence

$$\nabla_1 = 256(4B^3 - IA^2B - JA^3)\nabla.$$

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As a verification, suppose $(a, b, c, d, e)(x, y)^4 = x^4 + y^4$ (whence $I = 1, J = 0$). And take

$$x_1(x + y)^2 - y_1(x - y)^2 = 0$$

for the transforming equation, that is, $(\alpha, \beta, \gamma) = (1, 1, 1)$ and $(\alpha', \beta', \gamma') = (-1, 1, -1)$. We have $P = R = x_1 - y_1$ and $Q = x_1 + y_1$, and thence

$$\begin{aligned} \text{Det} &= (P^2 + R^2)^2 - 16PQ^2R + 16Q^4 \\ &= (2P^2 - 4Q^2)^2 = (-2x_1^2 - 12x_1y_1 - 2y_1^2)^2, \end{aligned}$$

that is,

$$(a_1, b_1, c_1, d_1, e_1)(x_1, y_1)^4 = 4(x_1^2 + 6x_1y_1 + y_1^2)^2,$$

whence

$$I_1 = \frac{4096}{3} = \frac{2^{12}}{3}, \quad J_1 = -\frac{262144}{27} = -\frac{2^{18}}{27};$$

also

$$A = -16, \quad B = 8,$$

and the equations for I_1, J_1 become

$$\begin{aligned} \frac{4096}{3} &= 4 \left(4 \cdot 64 + \frac{1}{3}256 \right), \\ -\frac{262144}{27} &= 8 \left(\frac{4}{3} \cdot (-16) \cdot 64 + \frac{1}{27}4096 \right), \end{aligned}$$

which are true. More generally, assuming

$$(a, b, c, d, e)(x, y)^4 = x^4 + 6\Theta x^2y^2 + y^4,$$

(whence $I = 1 + 3\Theta^2$, $J = \Theta - \Theta^3$), and the same transforming equation, we have

$$(a_1, b_1, c_1, d_1, e_1)(x_1, y_1)^4 = 4\{(1 + 3\Theta)x_1^2 + (3 - 3\Theta)2x_1y_1 + (1 + 3\Theta)y_1^2\}^2,$$

whence

$$I_1 = \frac{2^{12}}{3}(1 - 3\Theta)^2, \quad J_1 = -\frac{2^{18}}{27}(1 - 3\Theta)^3;$$

also

$$A = -16, \quad B = 8(1 - \Theta).$$

Substituting these different values in the equations for I_1, J_1 , we obtain

$$16(1 - 3\Theta)^2 = 12(1 + 3\Theta^2)(1 - \Theta)^2 - 72(\Theta - \Theta^3)(1 - \Theta) + 4(1 + 3\Theta^2)^2,$$

and

$$\begin{aligned} -8(1 - 3\Theta)^3 &= 27(\Theta - \Theta^3)(1 - \Theta)^3 - 9(1 + 3\Theta^2)^2(1 - \Theta)^2 \\ &\quad + 27(1 + 3\Theta^2)(\Theta - \Theta^3)(1 - \Theta) - 54(\Theta - \Theta^3)^2 + (1 + 3\Theta^2)^3, \end{aligned}$$

which are in fact satisfied identically.

Cambridge, 26 July, 1871.

[526] ON A SURFACE OF THE EIGHTH ORDER

[From the *Mathematische Annalen*, vol. IV. (1871), pp. 558–560.]

p. 401 [526; source PDF p. 465]

I reproduce in an altered form, so as to exhibit the application thereto of the theory of the six coordinates of a line, the analysis by which Dr Hierholzer obtained the equation of the surface of the eighth order, the locus of the vertex of a quadricone which touches six given lines.

I call to mind that if $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$ are the coordinates of any two points on a line, then the quantities (a, b, c, f, g, h) , which denote respectively

$$(\beta\gamma' - \beta'\gamma, \gamma\alpha' - \gamma'\alpha, \alpha\beta' - \alpha'\beta, \alpha\delta' - \alpha'\delta, \beta\delta' - \beta'\delta, \gamma\delta' - \gamma'\delta),$$

and which are such that $af + bg + ch = 0$, are the six coordinates of the line.¹

Consider the given point (x, y, z, w) and the given line (a, b, c, f, g, h) , and write for shortness

$$P = hy - gz + aw, \quad Q = -hx + fz + bw, \quad R = gx - fy + cw, \quad S = -ax - by - cz;$$

then taking (X, Y, Z, W) as current coordinates, the equation of the plane through the given point and line is

$$PX + QY + RZ + SW = 0.$$

Considering in like manner the given point (x, y, z, w) and the three given lines $(a_1, b_1, c_1, f_1, g_1, h_1)$, $(a_2, \dots)$, $(a_3, \dots)$, then we have the three planes

$$P_i X + Q_i Y + R_i Z + S_i W = 0, \quad i = 1, 2, 3,$$

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and if these planes have a common line, the point (x, y, z, w) is in a line meeting each of the three given lines; that is, the locus of the point is the hyperboloid through the three given lines. It follows that the equations

$$\begin{vmatrix} P_1 & Q_1 & R_1 & S_1 \\ P_2 & Q_2 & R_2 & S_2 \\ P_3 & Q_3 & R_3 & S_3 \end{vmatrix} = 0$$

reduce themselves to a single equation, that of the hyperboloid in question.

I write for shortness

$$\begin{aligned} (000) &= (agh)x^2 + (bhf)y^2 + (cfg)z^2 + (abc)w^2 \\ &\quad + [(abg) - (cah)]xw + [(bch) - (abf)]yw + [(caf) - (bcg)]zw \\ &\quad + [(bfg) + (chf)]yz + [(cgh) + (afg)]zx + [(ahf) + (bgh)]xy, \end{aligned}$$

viz. (123) will mean $(a_1 g_2 h_3)x^2 + \dots$, where $(a_1 g_2 h_3)$ etc. denote as usual the determinants

$$\begin{vmatrix} a_1 & g_1 & h_1 \\ a_2 & g_2 & h_2 \\ a_3 & g_3 & h_3 \end{vmatrix}, \quad \text{etc.}$$

Then the equations in question are found to be

$$x(123) = 0, \quad y(123) = 0, \quad z(123) = 0, \quad w(123) = 0,$$

¹Cayley, "On the six coordinates of a line," *Camb. Phil. Trans.* vol. XI. (1869), [435], pp. 290–323.

reducing themselves to the single equation $(123) = 0$, which is accordingly that of the hyperboloid through the three lines.²

Proceeding now to the above-mentioned problem, we have the point (x, y, z, w) , and the six lines $(a_1, b_1, c_1, f_1, g_1, h_1)$, $(a_2, \dots)$ etc., say the lines 1, 2, 3, 4, 5, 6: the six planes

$$P_1X + Q_1Y + R_1Z + S_1W = 0, \quad \text{etc.}$$

must be tangents to the same quadricone; that is, considering the sections by the plane $W = 0$, the six lines

$$P_1X + Q_1Y + R_1Z = 0, \quad \text{etc.}$$

must be tangents to the same conic, and the condition for this is

$$[123456] = 0,$$

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where the symbol stands for the determinant

$$\begin{vmatrix} P_1^2 & Q_1^2 & R_1^2 & Q_1R_1 & R_1P_1 & P_1Q_1 \\ P_2^2 & \dots & & & & \\ \vdots & & & & & \end{vmatrix}.$$

But as is well known this equation may be written

$$(*) \quad (126)(346)(145)(235) - (146)(236)(125)(345) = 0,$$

where (126) etc. denote the determinants

$$\begin{vmatrix} P_1 & Q_1 & R_1 \\ P_2 & Q_2 & R_2 \\ P_6 & Q_6 & R_6 \end{vmatrix}, \quad \text{etc.};$$

or, what is the same thing, they denote the functions above represented by the like symbols $(126) = (a_1g_2h_6)x^2 + \dots$. The equation $(*)$ just obtained is Hierholzer's equation for the surface of the eighth order, the locus of the vertex of a quadricone which touches six given lines.

I remark that in my "Memoir on Quartic Surfaces," *Proc. Lond. Math. Soc.* vol. III. (1870), [445], pp. 19–69, I obtained the equation of the surface under the foregoing form $[123456] = 0$ or say $[(P, Q, R)^2] = 0$, noticing that there was a factor w^4 , so that the order of the surface is = 8; and further that the equation might be written

$$w^8 \exp \frac{1}{w} \{x(g\partial_c - h\partial_b) + y(h\partial_a - f\partial_c) + z(f\partial_b - g\partial_a)\} [(a, b, c)^2] = 0,$$

where $\exp \Theta$ (read exponential) denotes e^Θ , and $[(a, b, c)^2]$ denotes

$$\begin{vmatrix} a_1^2 & b_1^2 & c_1^2 & b_1c_1 & c_1a_1 & a_1b_1 \\ a_2^2 & \dots & & & & \\ \vdots & & & & & \end{vmatrix}.$$

Also that the equation contains the four terms

$$x^8[(a, -h, g)^2] + y^8[(h, b, -f)^2] + z^8[(-g, f, c)^2] + w^8[(a, b, c)^2] = 0.$$

Cambridge, 12 September, 1871.

²This equation is given in the paper above referred to, §54.

[527] ON A THEOREM IN COVARIANTS

[From the *Mathematische Annalen*, vol. V. (1872), pp. 625–629.]

p. 404 [527; source PDF p. 468]

The proof given in Clebsch, *Theorie der binären algebraischen Formen* (Leipzig, 1872), of the finite number of the covariants of a binary form depends upon a subsidiary proposition which is deserving of attention for its own sake.

I use my own hyperdeterminant notation, which is as follows. Considering a function $U = (a, \dots)(x, y)^n$ (viz. $U_1 = (a, \dots)(x_1, y_1)^n$, &c.), and writing $12 = d_{x_1}d_{y_2} - d_{y_1}d_{x_2}$, &c., then the general form of a covariant of the degree m is

$$k(12^\alpha 23^\beta \dots)U_1U_2 \dots U_m,$$

where k is a merely numerical factor, the indices $\alpha, \beta, \gamma, \dots$ are positive integers, and after the differentiations each set of variables $(x_1, y_1), \dots, (x_m, y_m)$ is replaced by (x, y) . I say that the general form of a covariant is as above; viz. a covariant is equal to a single term of the above form, or a sum of such terms.

Attending to a single term: the sum of the indices of all the duads which contain a particular number $1, 2, \dots$ as the case may be is called an index-sum; each index-sum is at most $= n$; so that, calling the index-sums $\sigma_1, \sigma_2, \dots, \sigma_m$ respectively, we have $n - \sigma_1, n - \sigma_2, \dots, n - \sigma_m$ each of them zero or positive. The term, before the several sets of variables are each replaced by (x, y) , is of the orders $n - \sigma_1, n - \sigma_2, \dots, n - \sigma_m$ in the several sets of variables respectively.

The term may be expressed somewhat differently: for writing

$$V_1 = xd_{x_1} + yd_{y_1}, \quad V_2 = xd_{x_2} + yd_{y_2}, \quad \&c.,$$

then (except as to a numerical factor) it is for a function $\phi(x_1, y_1)$ the same thing whether we change (x_1, y_1) into (x, y) , or operate on this function with V_1^p , and so for the other sets; the term may therefore be written

$$V_1^{n-\sigma_1}V_2^{n-\sigma_2} \dots (12^\alpha 23^\beta \dots)U_1U_2 \dots U_m,$$

being now in the first instance a function of the single set (x, y) of variables.

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We may omit the operand $U_1U_2 \dots U_m$, and consider only the symbol

$$k(12^\alpha 13^\beta 23^\gamma \dots) \quad \text{or} \quad V_1^{n-\sigma_1} \dots V_m^{n-\sigma_m}(12^\alpha 13^\beta 23^\gamma \dots),$$

which, under either of the two forms, I represent for shortness by $[12 \dots m]$. Observe that this is considered as a symbol involving the m symbolic numbers $1, 2, 3, \dots, m$, even although in particular cases one or more of these numbers may be wanting from the actual expression of the symbol: thus $[123]$ may denote 12^n , but the operand to be supplied thereto is always $U_1U_2U_3$.

A sum of symbols is not in general equal to a single symbol; but a single symbol can be expressed in a variety of ways as a sum of symbols: the most simple transformation-formulae relate to three or four symbolic numbers; viz. for three such numbers, say $1, 2, 3$, we have

$$V_1.23 + V_2.31 + V_3.12 = 0,$$

showing that in a symbol, which written with the V 's involves $V_1.23$, this may be replaced by its value $V_2.13 - V_3.12$; and so in other cases.

For the four numbers 1, 2, 3, 4 we have a group of the like formulae

$$\begin{array}{rrrr} -V_{2.34} & +V_{3.24} & -V_{4.23} & = 0, \\ V_{1.34} & & -V_{3.14} & -V_{4.31} = 0, \\ -V_{1.24} & & +V_{3.14} & -V_{4.12} = 0, \\ V_{1.23} & +V_{2.31} & & = 0, \end{array}$$

leading to

$$23.14 + 31.24 + 12.34 = 0,$$

which is a form not involving the V 's and consequently is applicable to the transformation of invariant-symbols where the numbers

$$n - \sigma_1, n - \sigma_2, \dots, n - \sigma_m$$

are all = 0.

I establish the following definitions. A symbol $[12 \dots m]$ is proximate when each index-sum is $< n$; otherwise it is ultimate; viz. this is the case when any one or more of the index-sums is or are $= n$. We may say that the symbol is ultimate as to 1 if $\sigma_1 = n$; and that it is ultimate as to 1, 2 if σ_1 and σ_2 are each $= n$: and so in other cases.

A proximate symbol which has any one index-sum thereof $< \frac{1}{2}n$ is said to be inferior: thus if $\sigma_1 < \frac{1}{2}n$ the symbol is inferior in regard to 1; and so if σ_1 and σ_2 are each $< \frac{1}{2}n$, it is inferior in regard to 1 and 2: and the like in other cases.

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Observe that if a symbol is inferior then in the covariant the order exceeds the degree by a number which is greater than $\frac{1}{2}n - 1$: in fact, suppose it inferior in regard to 1, then the order is

$$(n - \sigma_1) + (n - \sigma_2) + \dots + (n - \sigma_m),$$

where each term after the first is at least = 1, that is, the order is at least $= n - \sigma_1 + m - 1$; hence order minus degree is at least $= n - \sigma_1 - 1$; viz. σ_1 being less than $\frac{1}{2}n$, this is greater than $\frac{1}{2}n - 1$. Conversely, if for any symbol order-degree is $> \frac{1}{2}n - 1$, then the symbol is not inferior.

A symbol $[12 \dots m]$ is sharp when any index is $> \frac{1}{2}n$; otherwise it is flat; viz. this is so when each index is $< \frac{1}{2}n$. The symbol is sharp as to any particular duad or duads when the index or indices thereof is or are each of them $> \frac{1}{2}n$.

The subsidiary theorem is now as follows: "A symbol is inferior or sharp: or it can be expressed as a sum of symbols each of which is inferior or sharp"—or what is the same thing, the only symbols which need to be considered are those which are either inferior or sharp.

Thus for the degree 1 the symbol is $[1]$ (which is simply unity), $\sigma_1 = 0$, and the symbol is inferior. For the degree 2 the symbol is $[2] = 12^k$; if $k < \frac{1}{2}n$ the symbol is inferior, if $k > \frac{1}{2}n$ then it is sharp.

A proof is first required for the degree 3; here

$$[123] = 12^\alpha 13^\beta 23^\gamma$$

($\beta + \gamma, \gamma + \alpha, \alpha + \beta$ each = or $< n$), which may very well be neither inferior nor sharp; for instance, if $n = 5$, we have $12^2 13^2 23^2$, where each index being = 2, the symbol is not sharp; and each index-sum being = 4, the symbol is not inferior. But writing the symbol in the form

$$V_1 V_2 V_3 12^2 13^2 23^2,$$

then by means of the relation

$$V_{1.23} + V_{2.31} + V_{3.12} = 0$$

(or, what is the same thing, $V_1.23 = V_2.13 - V_3.12$), the symbol becomes

$$\begin{aligned} & V_1 V_2 V_3 12^2 13^2 23^2 (V_2.13 - V_3.12), \\ & = V_2^2 V_3 12^2 13^3 23^2 - V_2 V_3^2 12^3 13^2 23^2, \end{aligned}$$

where each term, as containing an index 3, is sharp. To complete the reduction, observe that calling the expression $21 - 33$, then in the term 31 interchanging the numbers 2 and 3 we obtain $31 = -23$, and thence $21 - 23 = 221$; so that the whole is $2V_2^2 V_3 12^2 13^3 23^2$, viz. it is a multiple of a sharp term.

I prove the general case, substantially in the manner used by Dr Clebsch, as follows. We assume that the theorem is proved up to a particular degree m : that is, we assume that every symbol belonging to a degree not exceeding m can be

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expressed as a sum of terms each of which is sharp or inferior: and we have to prove this for the next following degree $m + 1$, or writing for convenience p in place of $m + 1$ (say for the degree p); that is, for a symbol

$$[12 \dots mp] = p1^{\lambda_1} p2^{\lambda_2} \dots pm^{\lambda_m} [12 \dots m] = P[12 \dots m], \quad \text{suppose.}$$

I write as before $\sigma_1, \sigma_2, \dots, \sigma_m$ for the index-sums of $[12 \dots m]$: those of $[12 \dots mp]$ are therefore $\sigma_1 + \lambda_1, \sigma_2 + \lambda_2, \dots, \sigma_m + \lambda_m$, and (for the duads involving p) $\sigma_p = \lambda_1 + \lambda_2 + \dots + \lambda_m$.

If $[12 \dots m]$ is sharp, then $[12 \dots mp]$ is sharp, and the theorem is true. If $\sigma_p < \frac{1}{2}n$, then $[12 \dots mp]$ is inferior in regard to p ; and the theorem is true. The only case requiring a proof is when $[12 \dots m]$ is not sharp (being therefore inferior) and when $\sigma_p > \frac{1}{2}n$. And in this case if any one of the indices $\lambda_1, \dots, \lambda_m$ is $> \frac{1}{2}n$ (or say if P is sharp) then the theorem is true.

Consider the expression

$$p1^{\lambda_1} p2^{\lambda_2} \dots pm^{\lambda_m} [12 \dots m],$$

where $\sigma_1, \sigma_2, \dots, \sigma_m$ are as before the index-sums for $[12 \dots m]$ and therefore the numbers

$$n - \sigma_1 - \lambda_1, \dots, n - \sigma_m - \lambda_m$$

are none of them negative.

Assume that when $[12 \dots m]$ is inferior, and when $\lambda_1, \dots, \lambda_m$ have any values such that their sum is not greater than a given value $\sigma_p - 1$, the expression is a sum of terms each of which is inferior or sharp: we wish to show that when $\lambda_1 + \lambda_2 + \dots + \lambda_m$ has the next succeeding value, $= \sigma_p$, the case is still the same.

For this purpose, introducing the V 's I write

$$Q = V_1^{n-\sigma_1-\lambda_1} \dots V_m^{n-\sigma_m-\lambda_m} V_p^{n-\sigma_p} p1^{\lambda_1} p2^{\lambda_2} \dots pm^{\lambda_m} [12 \dots m].$$

Then supposing for a moment that λ_1 is not $= n - \sigma_1$ and λ_2 is not $= 0$, the expression contains the factor $V_1.p2$, which is equal to and may be replaced by $-V_2.p1 + V_p.12$: we have thus, omitting the V 's,

$$\begin{aligned} Q &= p1^{\lambda_1+1} p2^{\lambda_2-1} p3^{\lambda_3} \dots pm^{\lambda_m} [12 \dots m], \\ Q' &= p1^{\lambda_1} p2^{\lambda_2-1} p3^{\lambda_3} \dots pm^{\lambda_m} 12[12 \dots m]. \end{aligned}$$

Now for Q' the sum of the indices $\lambda_1, \lambda_2 - 1, \lambda_3, \dots, \lambda_m$ is $\sigma_p - 1$, so that by hypothesis Q' is inferior or sharp: that is, the difference $Q - Q'$ is inferior or sharp; so that to prove that Q is inferior or sharp, we have only to prove this of Q , where this new Q is derived from the old one by increasing by unity the index of $p1$, at the expense of that of $p2$

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which is diminished by unity. Such change is possible so long as the index λ_1 has not attained its maximum value, $n - \sigma_1$ or σ_p as the case may be, and there is any other index $\lambda_2, \dots, \lambda_m$ which is not $= 0$: that is, we may pass from Q to Q' , from Q' to Q'' and so on; and it will be sufficient to show that the last term of the series is inferior or sharp. We thus pass from Q to R , where

$$R = p1^{n-\sigma_1}p2^{\lambda_2-\alpha_2} \dots pm^{\lambda_m-\alpha_m}[12 \dots m]$$

and $\alpha_2 + \alpha_3 + \dots + \alpha_m = n - \sigma_1 - \lambda_1$; or else to

$$R = p1^{\sigma_p}[12 \dots m],$$

according as $n - \sigma_1$ is not greater or is greater than σ_p .

Now let $[12 \dots m]$ be inferior; suppose it to be so in regard to 1, that is, let σ_1 be less than $\frac{1}{2}n$ or $n - \sigma_1$ greater than $\frac{1}{2}n$. Then if σ_p be less than $\frac{1}{2}n$ it is less than $n - \sigma_1$, that is, we have for R the last-mentioned form which is inferior in regard to p , viz. R is inferior; if σ_p is equal to or greater than $\frac{1}{2}n$, then R , whichever its form may be, is sharp as to $p1$, viz. R is sharp. Hence in either case Q is a sum of terms which are inferior or sharp; that is, assuming the theorem for a form for which $\lambda_1 + \lambda_2 + \dots + \lambda_m$ does not exceed a given value $\sigma_p - 1$, the theorem is true for the next succeeding value σ_p ; or being true for the case $\sigma_p - 1 = 0$, it is true generally.

Cambridge, 24 April, 1872.

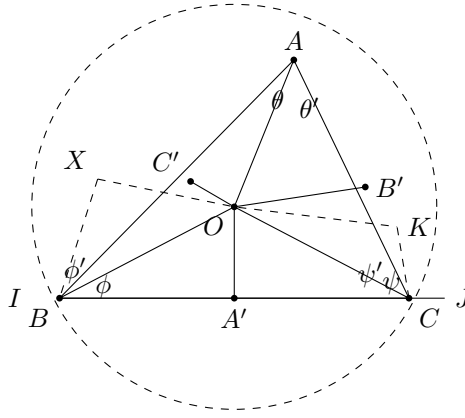
[528] ON THE NON-EUCLIDIAN GEOMETRY

[From the *Mathematische Annalen*, vol. V. (1872), pp. 630–634.]

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The theory of the Non-Euclidian Geometry as developed in Dr Klein's paper "Ueber die Nicht-Euklidische Geometrie" may be illustrated by showing how in such a system we actually measure a distance and an angle and by establishing the trigonometry of such a system. I confine myself to the "hyperbolic" case of plane geometry; viz. the absolute is here a real conic, which for simplicity I take to be a circle; and I attend to the points *within* the circle.

I use the simple letters $a, A, \dots$ to denote (linear or angular) distances measured in the ordinary manner; and the same letters, with a superscript stroke, $\bar{a}, \bar{A}, \dots$, to denote the same distances measured according to the theory.



The radius of the absolute is for convenience taken to be $= 1$; the distance of any point from the centre can therefore be represented as the sine of an angle.

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The distance BC , or say $\bar{a}$, of any two points B, C is by definition as follows:

$$\begin{aligned} \text{Radius of circle} &= 1, \\ \text{in } \triangle ABC, \text{ sides are} & a, b, c, \\ \text{angles} & A, B, C, \\ OA, OB, OC &= \sin p, \sin q, \sin r, \\ OA', OB', OC' &= \sin a, \sin b, \sin c, \\ \angle BOC, \angle COA, \angle AOB &= \alpha, \beta, \gamma. \end{aligned}$$

Where I, J are the intersections of the line BC with the circle,

$$e^{\bar{a}} + e^{-\bar{a}}, \quad \text{or} \quad 2 \cosh \bar{a} = \frac{BI.CJ}{BJ.CI} + \frac{BJ.CI}{BI.CJ} = \frac{BI.CJ + BJ.CI}{\sqrt{BI.BJ.CI.CJ}}.$$

The numerator is

$$BI(BJ - BC) + CI(BC + CJ) = BI.BJ + CI.CJ + BC(CI - BI) = BI.BJ + CI.CJ + BC^2.$$

Hence taking a for the distance BC , and $\sin q, \sin r$ for the distances OB, OC respectively, we have $BI.BJ = \cos^2 q$, $CI.CJ = \cos^2 r$; and the formula is

$$\cosh \bar{a} = \frac{\cos^2 q + \cos^2 r + a^2}{2 \cos q \cos r}.$$

Or, what is the same thing, taking α for the angle BOC , and therefore

$$a^2 = \sin^2 q + \sin^2 r - 2 \sin q \sin r \cos \alpha,$$

we have

$$\cosh \bar{a} = \frac{1 - \sin q \sin r \cos \alpha}{\cos q \cos r}.$$

In a similar manner, if $\sin a$ is the perpendicular distance from O on the line BC (that is, $a \sin a = \sin q \sin r \sin \alpha$), it can be shown that

$$\sinh \bar{a} = \frac{a \cos a}{\cos q \cos r},$$

the equivalence of the two formulae appearing from the identity

$$\cos^2 q \cos^2 r = (1 - \sin q \sin r \cos \alpha)^2 - a^2 + a^2 \sin^2 a,$$

which is at once verified.

Next for an angle; we have by definition

$$\bar{A} = \frac{1}{2i} \log \frac{\sin BAI \cdot \sin CAJ}{\sin BAJ \cdot \sin CAI}.$$

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Where AI, AJ are the (imaginary) tangents from A to the circle; or writing for shortness BI &c. instead of BAI &c. (the angular point being always at A),

$$\bar{A} = \frac{1}{2i} \log \frac{\sin BI \sin CJ}{\sin BJ \sin CI}.$$

Consequently

$$e^{i\bar{A}} - e^{-i\bar{A}} = 2i \sin \bar{A} = \frac{\sin BI \sin CJ - \sin BJ \sin CI}{\sqrt{\sin BI \sin CJ} \sqrt{\sin BJ \sin CI}},$$

where the numerator is

$$\sin BI \sin(CJ - BC) - \sin CJ \sin(BI + BC) = \sin BC \sin IJ,$$

or say $= \sin A \sin IJ$. Moreover taking the distance OA to be $= \sin p$, and the perpendicular distances from O on the lines AB, AC to be $\sin c$ and $\sin b$ respectively, then if for a moment the angle IJ is put $= 2\omega$, we have $\sin p \sin \omega = 1$; moreover

$$\sin BI \sin BJ = \sin(\omega - BO) \sin(\omega + BO) = \sin^2 \omega - \sin^2 BO,$$

and $\sin p \sin BO = \sin c$; that is,

$$\sin BI \sin BJ = \frac{\cos^2 c}{\sin^2 p}, \quad \sin CI \sin CJ = \frac{\cos^2 b}{\sin^2 p}.$$

Also

$$\sin IJ = -\sin 2\omega = 2 \sin \omega \cos \omega = \frac{2 \cos p}{\sin^2 p}.$$

Whence the required formula

$$\sin \bar{A} = \frac{\cos p \sin A}{\cos b \cos c}.$$

In the same way, or analytically from this value, we have

$$\cos \bar{A} = \frac{\cos A + \sin b \sin c}{\cos b \cos c},$$

and thence also

$$\tan \bar{A} = \frac{\cos p \sin A}{\cos A + \sin b \sin c}.$$

In particular, taking the line AC to pass through O , or writing in the formula $b = 0$, we have

$$BO = \tan^{-1}(\cos p \tan \theta), \quad CO = \tan^{-1}(\cos p \tan \theta');$$

we ought to have

$$A = BO + CO = \tan^{-1}(\cos p \tan \theta) + \tan^{-1}(\cos p \tan \theta'),$$

which, observing that $\sin p \sin \theta = \sin c$ and $\sin p \sin \theta' = \sin b$, also $A = \theta + \theta'$, is in fact equivalent to the above formula for $\tan \bar{A}$. Observe in particular that when A is at the centre, $p = 0$, and the formula becomes $\bar{A} = \theta + \theta' = A$, or say for an angle at the centre, $\bar{A} = A$.

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I return to the expression for $\cosh \bar{a}$; in explanation of its meaning, let the distances OB, OC be q, r respectively and let the angle BOC be α ; to find $\bar{q}$ we have only to take C at O , that is, in the formula for $\cosh \bar{a}$ to write $r = 0$, we thus find

$$\cosh \bar{q} = \frac{1}{\cos q}, \quad \cosh \bar{r} = \frac{1}{\cos r},$$

whence also

$$\cos q = \operatorname{sech} \bar{q}, \quad \sin q = i \tanh \bar{q}, \quad \cos r = \operatorname{sech} \bar{r}, \quad \sin r = i \tanh \bar{r}.$$

Also, as seen above, $\alpha = \alpha$; the formula thus is

$$\begin{aligned} \cosh \bar{a} &= \frac{1 + \tanh \bar{q} \tanh \bar{r} \cos \alpha}{\operatorname{sech} \bar{q} \operatorname{sech} \bar{r}} \\ &= \cosh \bar{q} \cosh \bar{r} + \sinh \bar{q} \sinh \bar{r} \cos \alpha, \end{aligned}$$

or, what is the same thing, it is

$$\cos \alpha = \frac{\cosh \bar{a} - \cosh \bar{q} \cosh \bar{r}}{\sinh \bar{q} \sinh \bar{r}},$$

viz. as will presently appear, this is the formula for $\cos BOC$ in the triangle BOC .

From the above formulae

$$\cosh \bar{a} = \frac{1 - \sin q \sin r \cos \alpha}{\cos q \cos r},$$

and

$$\sin \bar{A} = \frac{\cos p \sin A}{\cos b \cos c}, \quad \cos \bar{A} = \frac{\cos A + \sin b \sin c}{\cos b \cos c},$$

and the like formulae for b, c, B, C , it may be shown that in the triangle ABC we have

$$\cosh \bar{a} = \frac{\cos \bar{A} + \cos \bar{B} \cos \bar{C}}{\sin \bar{B} \sin \bar{C}}.$$

In fact, substituting the foregoing values, this equation becomes

$$\frac{(1 - \sin^2 a)(\cos A + \sin b \sin c) + (\cos B + \sin c \sin a)(\cos C + \sin a \sin b)}{\sin B \sin C \cos b \cos c} = \frac{1 - \sin q \sin r \cos \alpha}{\cos q \cos r},$$

that is,

$$\begin{aligned} \cos A + \cos B \cos C - \sin^2 a \cos A + \sin a \sin b \cos B \\ + \sin a \sin c \cos C + \sin b \sin c = \sin B \sin C (1 - \sin q \sin r \cos \alpha), \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} & \sin^2 a (\cos B \cos C - \sin B \sin C) + \sin a \sin b \cos B \\ & + \sin a \sin c \cos C + \sin b \sin c = -\sin B \sin C \sin q \sin r \cos \alpha, \end{aligned}$$

that is,

$$(\sin a \cos B + \sin c)(\sin a \cos C + \sin b) = \sin B \sin C (\sin^2 a - \sin q \sin r \cos \alpha),$$

a relation which I proceed to verify.

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We may, from the formulae

$$a^2 = \sin^2 q + \sin^2 r - 2 \sin q \sin r \cos \alpha, \quad a \sin a = \sin q \sin r \sin \alpha, \quad \&c.,$$

but, more simply, geometrically as presently shown, deduce

$$\sin a \cos B + \sin c = a^{-1} \sin B \sin q (\sin q - \sin r \cos \alpha),$$

$$\sin a \cos C + \sin b = a^{-1} \sin C \sin r (\sin r - \sin q \cos \alpha),$$

and thence

$$\begin{aligned} & (\sin a \cos B + \sin c)(\sin a \cos C + \sin b) \\ & = a^{-2} \sin B \sin C \sin q \sin r \{ \sin q \sin r (1 + \cos^2 \alpha) - \cos \alpha (\sin^2 q + \sin^2 r) \} \\ & = \sin B \sin C (\sin^2 a - \sin q \sin r \cos \alpha), \end{aligned}$$

which is the equation in question. For the subsidiary equations used in the demonstration, observe that the four points O, X, A', B lie in a circle, and consequently that

$$CO.CX = CA'.CB;$$

or multiplying each side by $\sin C$, then

$$CO.CX. \sin C = A'.K.CB,$$

that is,

$$\sin r (\sin r - \sin q \cos \alpha) \sin C = a (\sin a \cos C + \sin b),$$

and the other of the equations in question is proved in the same manner.

From the formula for $\cosh \bar{a}$ we find

$$\sinh \bar{a} = \frac{\Delta}{\sin B \sin C},$$

where

$$\Delta^2 = -(1 - \cos^2 A - \cos^2 B - \cos^2 C - 2 \cos A \cos B \cos C),$$

whence also

$$\sinh \bar{a} : \sinh \bar{b} : \sinh \bar{c} = \sin A : \sin B : \sin C;$$

and we can also obtain

$$\cos \bar{A} = \frac{\cosh \bar{a} - \cosh \bar{b} \cosh \bar{c}}{\sinh \bar{b} \sinh \bar{c}}.$$

So that the formulae are in fact similar to those of spherical trigonometry with only $\cosh \bar{a}, \sinh \bar{a}$ &c. instead of $\cos a, \sin a$ &c. The before-mentioned formula for $\cos \alpha$ in terms of $\bar{a}, \bar{q}, \bar{r}$ is obviously a particular case of the last-mentioned formula for $\cos \bar{A}$.

Cambridge, 11 May, 1872.

[529] A “SMITH’S PRIZE” PAPER; SOLUTIONS

[From the *Oxford, Cambridge and Dublin Messenger of Mathematics*, vol. IV. (1868), pp. 201–226.]

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1. Find the form of a function of a given number of letters, which has two and only two values.

It is required to find the general form of a function $\phi(a, b, c, \dots, k)$, rational and integral, which for all permutations whatever of the letters has two and only two values.

Suppose that any particular permutation of the letters changes $\phi(a, b, c, \dots, k)$ into $\phi_1(a, b, c, \dots, k)$; then any permutation of the letters will either leave the functions ϕ, ϕ_1 each of them unaltered, or it will change ϕ into ϕ_1 , and ϕ_1 into ϕ . Hence $\phi + \phi_1$ is a symmetrical function of all the letters, say

$$\phi + \phi_1 = 2L;$$

$\phi - \phi_1$ is a function which by any permutation of the letters is either unaltered, or simply changes its sign; and it is to be shown that, writing for shortness V to denote the product

$$(a - b)(a - c) \cdots (b - c) \cdots$$

of the differences of the letters, and denoting by $2M$ a symmetrical function of the letters, we have

$$\phi - \phi_1 = 2VM.$$

These equations give

$$\phi = L + VM,$$

which is the general form required; viz. the function ϕ has then only the two values $L + VM, L - VM$.

To prove the subsidiary theorem, observe that there is at least one interchange of two letters which changes $\phi - \phi_1$ (for otherwise $\phi - \phi_1$ would be a symmetrical function).³

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Let this be (a, b) . Then $\phi - \phi_1$, changing its sign for the interchange in question, must vanish for $a = b$, that is, $\phi - \phi_1$ must contain the factor $(a - b)$; let it contain it in the power $(a - b)^s$; then the quotient $\phi - \phi_1 \div (a - b)^s$, not vanishing for $a = b$, cannot change its sign by the interchange in question, and as by supposition $\phi - \phi_1$ does change its sign, it appears that the exponent s must be odd. But $\phi - \phi_1$, containing the factor $(a - b)^s$, must contain every other like factor $(a - c)^s$ or $(c - d)^s$; in fact, writing $\phi - \phi_1 = K(a - b)^s$, then if the interchange (a, c) alters K into K_1 , we have

$$K(a - b)^s = \pm K_1(a - c)^s;$$

and K consequently contains the factor $(a - c)^s$; it does not contain any higher power $(a - c)^{s+s'}$, for if it did, by reversing the process it would appear that $\phi - \phi_1$ contained (contrary to supposition) the factor $(a - b)^{s+s'}$. Similarly $\phi - \phi_1$ contains the factor $(c - d)^s$, but no higher power $(c - d)^{s+s'}$. Hence $\phi - \phi_1$ contains the product of all the factors $(a - b)^s$, that is, it contains V^s , and writing

$$\phi - \phi_1 = 2V^s M,$$

the quotient M does not contain any such factor as $(a - b)$; it therefore does not change its sign for any interchange whatever (a, b) ; and in consequence it remains unaltered for any such interchange,

³Set by me, for the Master of Trinity, Thursday, January 30, 1868.

that is, M is a symmetrical function. Observing that any even power of V is a symmetrical function, we may without loss of generality include V^{s-1} in the symmetrical function M , and write therefore

$$\phi - \phi_1 = 2VM,$$

which is the subsidiary theorem in question.

2. Express $\frac{x^7 - 1}{x - 1}$ as the product of two cubic factors; and show generally that, for any prime exponent p , the function $\frac{x^p - 1}{x - 1}$ may be broken up into two factors each of the order $\frac{1}{2}(p - 1)$, by means of a quadratic equation.

Let r be any root of the given equation, then

$$r^6 + r^5 + r^4 + r^3 + r^2 + r + 1 = 0;$$

the roots of this equation are $r^1, r^2, r^3, r^4, r^5, r^6$. Hence

$$x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = (x - r^1)(x - r^2)(x - r^4)(x - r^3)(x - r^5)(x - r^6),$$

and denoting the two cubic factors by y_1, y_2 , or writing

$$y_1 = (x - r^1)(x - r^2)(x - r^4), \quad y_2 = (x - r^3)(x - r^5)(x - r^6),$$

we have

$$\begin{aligned} y_1 &= x^3 - x^2(r^1 + r^2 + r^4) + x(r^3 + r^5 + r^6) - 1, \\ y_2 &= x^3 - x^2(r^3 + r^5 + r^6) + x(r^1 + r^2 + r^4) - 1, \end{aligned}$$

and thence

$$y_1 + y_2 = 2x^3 + x^2 - x - 2.$$

Hence y_1, y_2 are the roots of the quadratic equation

$$y^2 - y(2x^3 + x^2 - x - 2) + x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = 0.$$

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Solving the equation, observing that the quantity under the radical sign must of necessity be a square, and that its value is

$$(2x^3 + x^2 - x - 2)^2 - 4(x^6 + x^5 + x^4 + x^3 + x^2 + x + 1),$$

which is in fact

$$= -7(x^2 + x)^2,$$

we find that the roots y_1, y_2 , that is, the required cubic factors, are

$$\frac{1}{2}\{2x^3 + x^2 - x - 2 \pm \sqrt{-7}(x^2 + x)\}.$$

Generally for any prime exponent p , denoting any root by r , and writing

$$y_1 = (x - r^{a_1})(x - r^{a_2}) \cdots, \quad y_2 = (x - r^{b_1})(x - r^{b_2}) \cdots,$$

where $a_1, a_2, \dots$ are the quadratic residues, and $b_1, b_2, \dots$ the quadratic non-residues of p , we find

$$y^2 - yP + Q = 0,$$

where P is a function of x of the order $\frac{1}{2}(p - 1)$, and

$$Q = x^{p-1} + x^{p-2} + \cdots + x + 1.$$

Hence

$$y = \frac{1}{2}\{P \pm \sqrt{P^2 - 4Q}\},$$

where $P^2 - 4Q$ is a perfect square, $= \varepsilon p Z^2$, Z a rational function of a degree less than $\frac{1}{2}(p-1)$, and $\varepsilon = +$ or $-$ according as $p \equiv 1$ or $\equiv 3 \pmod{4}$. Hence the required factors are

$$\frac{1}{2}\{P \pm \sqrt{\varepsilon p} Z\}.$$

3. *In a Map of the World, wherein the meridians are projected into right lines meeting in a point, the inclination of any two of the lines being equal to that of the two meridians, and the parallels into circles about the point as centre, the radius of the circle being a given function of the colatitude: compare on the sphere and in the map (1) the corresponding elements of area, (2) the azimuths of corresponding linear elements; and explain what conveniences may be obtained by proper determinations of the above-mentioned function of the colatitude.*

Let the longitude and colatitude be

$$\text{on the sphere } l, c, \quad \text{in the map } l', c',$$

then the projection is such, that $l' = l$, $c' = f(c)$, a given function of c . The lengths of corresponding linear elements in the direction of a meridian, and perpendicular to it, are

$$dc, \quad \sin c \, dl,$$

$$dc', \quad c' \, dl.$$